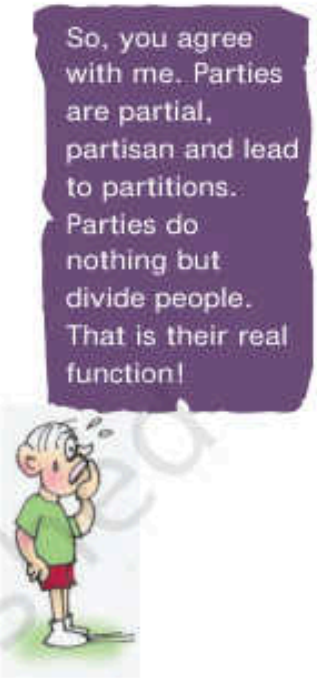
Why do we need political parties?

Political parties are easily one of the most visible institutions in a democracy. For most ordinary citizens, democracy is equal to political parties. If you travel to remote parts of our country and speak to the less educated citizens, you could come across people who may not know anything about our Constitution or about the nature of our government. But chances are that they would know something about our political parties. At the same time, this visibility does not mean popularity. Most people tend to be very critical of political parties. They tend to blame parties for all that is wrong with our democracy and our political life. Parties have become identified with social and political divisions.

Therefore, it is natural to ask – do we need political parties at all? About hundred years ago, there were few countries of the world that had any political party. Now there are few that do not have parties. Why did political parties become so omnipresent in democracies all over the world? Let us first answer what political parties are and what they do, before we say why we need them.



Meaning

A political party is a group of people who come together to contest elections and hold power in the government. They agree on some policies and programmes for the society with a view to promote the collective good. Since there can be different views on what is good for


(1)


(2)


(3)


(4)

Election Commission has officially banned wall writing by parties during election times. Most political parties argue that was the cheapest way for their campaign. These election times used to create amazing graffiti on the walls. Here are some examples from Tamil Nadu.

© (1) M Govarthan (2) A Muralidharan (3) M Moorthy (4) T Singaravelou, *The Hindu*



© RK Laxman - *Brushing up the years*

all, parties try to persuade people why their policies are better than others. They seek to implement these policies by winning popular support through elections.

Thus, parties reflect fundamental political divisions in a society. Parties are about a part of the society and thus, involve PARTISANSHIP. Thus, a party is known by which part it stands for, which policies it supports and whose interests it upholds. A political party has three components:

- the leaders,
- the active members and
- the followers

Functions

What does a political party do? Basically, political parties fill political offices and exercise political power. Parties do so by performing a series of functions:

1 Parties **contest elections**. In most democracies, elections are fought mainly among the candidates put up by political parties. Parties select their candidates in different ways. In some countries, such as the USA, members and supporters of a party choose its candidates. Now more and more countries are following this method. In other countries like India, top party leaders choose candidates for contesting elections.

2 Parties put forward different **policies and programmes** and the voters choose from them. Each of us may have different opinions and views on what policies are suitable for the society. But no government can handle such a large variety of views. In a democracy, a large number of similar opinions have to be grouped together to provide a direction in which policies can be formulated by the governments. This is what the parties do. A party reduces a vast multitude of opinions into a few basic positions which it

Glossary

Partisan: A person who is strongly committed to a party, group or faction. Partisanship is marked by a tendency to take a side and inability to take a balanced view on an issue.

Reprint 2026-27

supports. A government is expected to base its policies on the line taken by the RULING PARTY.

3 Parties play a decisive role in **making laws** for a country. Formally, laws are debated and passed in the legislature. But since most of the members belong to a party, they go by the direction of the party leadership, irrespective of their personal opinions.

4 Parties **form and run governments.** As we noted last year, the big policy decisions are taken by political executive that comes from the political parties. Parties recruit leaders, train them and then make them ministers to run the government in the way they want.

5 Those parties that lose in the elections play the **role of opposition** to the parties in power, by voicing different views and criticising government for its failures or wrong policies. Opposition parties also mobilise opposition to the government.

6 Parties **shape public opinion.** They raise and highlight issues. Parties have lakhs of members and activists spread all over the country. Many of the pressure groups are the extensions of political parties among different sections of society. Parties sometimes also launch movements for the resolution of problems faced by people. Often opinions in the society crystallise on the lines parties take.

7 Parties provide people **access to government machinery and welfare schemes** implemented by governments. For an ordinary citizen it is easy to approach a local party leader than a government officer. That is why, they feel close to parties even when they do not fully trust them. Parties have to be responsive to people's needs and demands. Otherwise people can reject those parties in the next elections.

Necessity

This list of functions in a sense answers the question asked above: we need political parties because they perform all these functions. But we still need to ask why modern democracies cannot exist without political parties. We can understand the necessity of political parties by imagining a situation without parties. Every candidate in the elections will be independent. So no one will be able to make any promises to the people about any major policy changes. The government may be formed, but its utility will remain ever uncertain. Elected representatives will be accountable to their constituency for what they do in the locality. But no one will be responsible for how the country will be run.

We can also think about it by looking at the non-party based elections to the panchayat in many states. Although, the parties do not contest formally, it is generally noticed that the village gets split into more than one faction, each of which puts up a 'panel' of its candidates. This is exactly what the party does. That is the reason we find political parties in almost all countries of the world, whether these countries are big or small, old or new, developed or developing.

The rise of political parties is directly linked to the emergence of representative democracies.



Okay, granted that we can't live without political parties. But tell me on what grounds do people support a political party?

Glossary

Ruling Party: Political party that runs government.

As we have seen, large societies need representative democracy. As societies became large and complex, they also needed some agency to gather different views on various issues and to present these to the government. They needed some ways, to bring various representatives together so that a responsible government could be formed. They needed a mechanism to support or restrain the government, make policies, justify or oppose them. Political parties fulfill these needs that every representative government has. We can say that parties are a necessary condition for a democracy.



Categorise these photographs by the functions of political parties they illustrate. Find one photograph or news clipping from your own area for each of the functions listed above.



© (1) C V Subrahmanyan (2) K Gopinathan (3) A Chakrabarty, *The Hindu*

1: Activists of BJP Mahila Morcha demonstrate against hike in prices of onions and LPG in Visakhapatnam.
2: Minister distributes ₹ One lakh cheque to the families of hooch victims at their houses.
3: Activists of CPI (M), CPI, OGP and JD (S) take out a rally in Bhubaneswar to protest against POSCO, the Korean steel company for being permitted by the State Government to export iron ore from Orissa to feed steel plants in China and Korea.

How many parties should we have?

In a democracy any group of citizens is free to form a political party. In this formal sense, there are a large number of political parties in each country. More than 750 parties are registered with the Election Commission of India. But not all these parties are serious contenders in the elections. Usually only a handful of parties are effectively in the race to win elections and form the government. So the question is: how many major or effective parties are good for a democracy?

In some countries, only one party is allowed to control and run the government. These are called one-party systems. In Class IX, we noted that in China, only the Communist Party is allowed to

Reprint 2026-27

rule. Although, legally speaking, people are free to form political parties, it does not happen because the electoral system does not permit free competition for power. We cannot consider one-party system as a good option because this is not a democratic option. Any democratic system must allow at least two parties to compete in elections and provide a fair chance for the competing parties to come to power.

In some countries, power usually changes between two main parties. Several other parties may exist, contest elections and win a few seats in the national legislatures. But only the two main parties have a serious chance of winning majority of seats to form government. Such a party system is called two-party system. The United States of America and the United Kingdom are examples of two-party system.

If several parties compete for power, and more than two parties have a reasonable chance of coming to power either on their own strength or in alliance with others, we call it a multiparty system. Thus in India, we have a multiparty system. In this system, the government is formed by various parties coming together in a coalition. When several parties in a multi-party system join hands for the purpose of contesting elections and winning power, it is called an alliance or a front. For example, in India there were three such major alliances in 2004 parliamentary elections– the National Democratic Alliance, the United Progressive Alliance and the Left Front. The multiparty system often appears very messy and leads to political instability. At the same time, this system allows a variety of interests and opinions to enjoy political representation.

So, which of these is better? Perhaps the best answer to this very common question is that this is not a very good question. Party system is not something any country can choose. It evolves over a long time, depending on the nature of society, its social and regional divisions, its history of politics and its system of elections. These cannot be changed very quickly. Each country develops a party system that is conditioned by its special circumstances. For example, if India has evolved a multiparty system, it is because the social and geographical diversity in such a large country is not easily absorbed by two or even three parties. No system is ideal for all countries and all situations.



I wonder how politicians manage these coalitions. I can't even remember the names of all the parties.



Let us apply what we have learnt about party systems to the various states within India. Here are three major types of party systems that exist at the State level. Can you find the names of at least two States for each of these types?

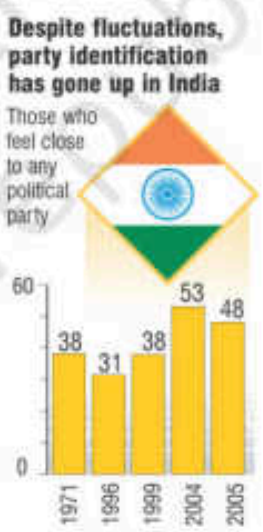
- Two-party system
- Multiparty system with two alliances
- Multiparty system

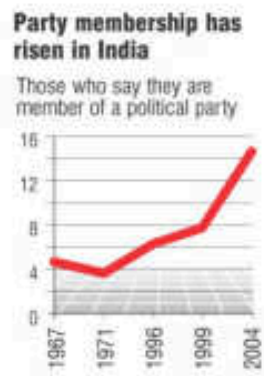
Popular participation in political parties

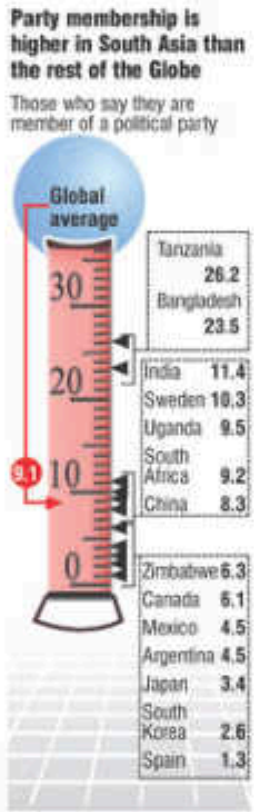
It is often said that political parties are facing a crisis because they are very unpopular and the citizens are indifferent to political parties. The available evidence shows that this belief is only partly true for India. The evidence, based on a series of large sample surveys conducted over several decades, shows that:

- Political parties do not enjoy much trust among the people in South Asia. The proportion of those who say their trust in political parties is 'not much' or 'not at all' is more than those who have 'some' or 'great' trust.
- The same is true of most other democracies as well. Political parties are one of the least trusted institutions all over the world.
- Yet the level of participation in the activities of political parties was fairly high. The proportion of those who said they were members of some political party was higher in India than many advanced countries like Canada, Japan, Spain and South Korea.
- Over the last three decades, the proportion of those who report to be members of political parties in India has gone up steadily.
- The proportion of those who say they feel 'close to a political party' has also gone up in India in this period.



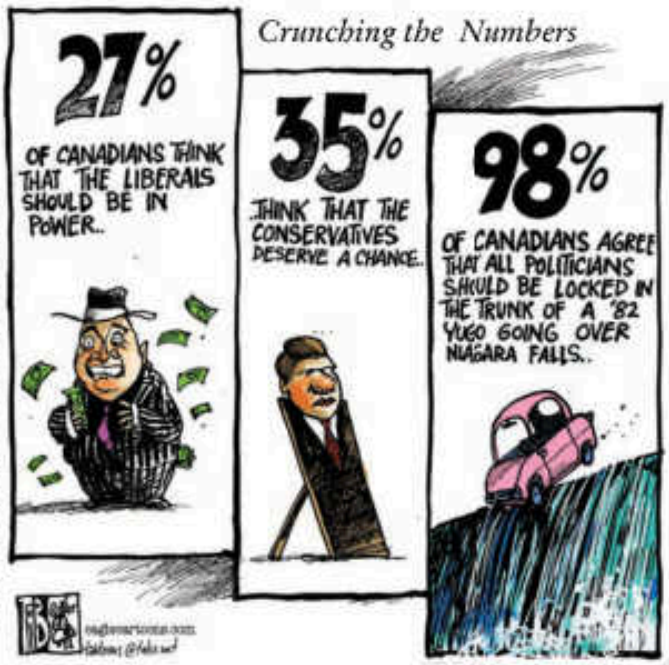






Source: SDSA Team, *State of Democracy in South Asia*, Delhi: Oxford University Press, 2007

© NCERT not to be republished



© Tab - *The Calgary Sun*, Cagle Cartoons Inc.

Does the cartoon reflect the data graphics shown on the previous page?

National parties

Democracies that follow a federal system all over the world tend to have two kinds of political parties: parties that are present in only one of the federal units and parties that are present in several or all units of the federation. This is the case in India as well. There are some country-wide parties, which are called 'national parties'. These parties have their units in various states. But by and large, all these units follow the same policies, programmes and strategy that is decided at the national level.

Every party in the country has to register with the Election Commission. While the Commission treats all parties equally, it offers some special facilities to large and established parties. These parties are given a unique symbol – only the official candidates of that party can use that election symbol. Parties that get this privilege and some other special facilities are 'recognised' by the Election Commission for this purpose. That is why these parties are called, 'recognised political parties'. The Election Commission has laid down detailed criteria of the proportion of votes and seats that a party must get in order to be a recognised party. A party that secures at least six per cent of the total votes in an election to the Legislative Assembly of a State and wins at least two seats is recognised as a State party. A party that secures at least six per cent of the total votes in Lok Sabha elections or Assembly elections in four States and wins at least four seats in the Lok Sabha is recognised as a national party.

According to this classification, there are six recognized national parties in the country as per notification of the Election Commission of India issued in 2023.

For more details about registration and recognition of political parties by the Election Commission of India, visit https://eci.gov.in

© NCERT not to be republished

Challenges to political parties

We have seen how crucial political parties are for the working of democracy. Since parties are the most visible face of democracy, it is natural that people blame parties for whatever is wrong with the working of democracy. All over the world, people express strong dissatisfaction with the failure of political parties to perform their functions well. This is the case in our country too. Popular dissatisfaction and criticism has focussed on four problem areas in the working of political parties. Political parties need to face and overcome these challenges in order to remain effective instruments of democracy.

The first challenge is **lack of internal democracy** within parties. All over the world there is a tendency in political parties towards the concentration of power in one or few leaders at the top. Parties do not keep membership registers, do not hold organisational meetings, and do not conduct internal elections regularly. Ordinary members of the party do not get sufficient information on what happens inside the party. They do not have the means or the connections needed to influence the decisions. As a result, the leaders assume greater power to make decisions in the name of the party. Since one or few leaders exercise paramount power in the party, those who disagree with the leadership find

Berlusconi Puppet Theatre



© Riber Hansson - *Svenska Dagbladet*, Cagle Cartoons Inc.

Berlusconi was the Prime Minister of Italy. He is also one of the top businessmen in Italy. He is the leader of the Forza Italia founded in 1993. His company owns TV channels, the most important publishing company, a foot ball club (AC Milan) and a bank. This cartoon was made during the last elections.

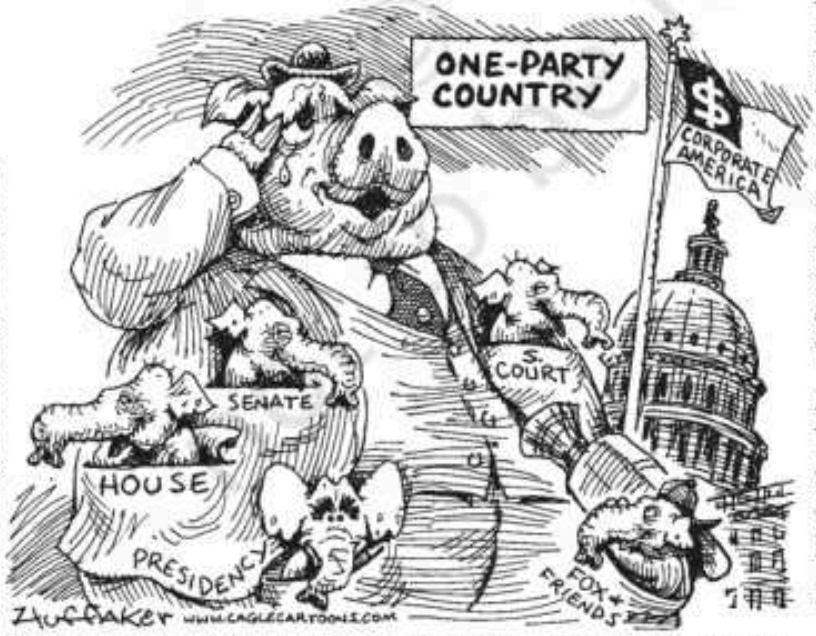


Why don't parties give enough tickets to women? Is that also due to lack of internal democracy?

© NCERT not to be republished

it difficult to continue in the party. More than loyalty to party principles and policies, personal loyalty to the leader becomes more important.

The second challenge of dynastic succession is related to the first one. Since most political parties do not practice open and transparent procedures for their functioning, there are very few ways for an ordinary worker to rise to the top in a party. Those who happen to be the leaders are in a position of unfair advantage to favour people close to them or even their family members. In many parties, the top positions are always controlled by members of one family. This is unfair to other members of that party. This is also bad for democracy, since people who do not have adequate experience or popular support come to occupy positions of power. This tendency is present in some measure all over the world, including in some of the older democracies.



This cartoon was drawn during the Presidency of George Bush of the Republican Party in the USA. The party's symbol is elephant. The cartoon seems to suggest that the Corporate America controls all major institutions of the country.

The third challenge is about the growing role of **money and muscle power** in parties, especially during elections. Since parties are focussed only on winning elections, they tend to use short-cuts to win elections. They tend to nominate those candidates who have or can raise lots of money. Rich people and companies who give funds to the parties tend to have influence on the policies and decisions of the party. In some cases, parties support criminals who can win elections. Democrats all over the world are worried about the increasing role of rich people and big companies in democratic politics.

The fourth challenge is that very often parties do not seem to offer a **meaningful choice** to the voters. In order to offer meaningful choice, parties must be significantly different. In recent years, there has been a decline in the ideological differences among parties in most parts of the world. For example, the difference between the Labour Party and the Conservative Party in Britain is very little. They agree on more fundamental aspects but differ only in details on how policies are to be framed and implemented. In our country too, the differences among all the major parties on the economic policies have reduced. Those who want really different policies have no option available to them. Sometimes people cannot even elect very different leaders either, because the same set of leaders keep shifting from one party to another.

Reprint 2026-27



You already have so much money. Why do you want to contest the elections?

© Manjul - DNA



Does this suggest that in democracies people contest elections only to make money? But isn't it true that there are politicians committed to the well-being of the people?



Can you identify which of the challenges described in this section are being highlighted in these cartoons (on pages 57 to 59)? What are the ways to curb the misuse of money and muscle power in politics?

How can parties be reformed?

In order to face these challenges, political parties need to be reformed. The question is: Are political parties willing to reform? If they are willing, what has prevented them from reforming so far? If they are not willing, is it possible to force them to reform? Citizens all over the world face this question. This is not a simple question to answer. In a democracy, the final decision is made by leaders who represent political parties. People can replace them, but only by another set of party leaders. If all of them do not wish to reform, how can anyone force them to change?

Let us look at some of the recent efforts and suggestions in our country to reform political parties and its leaders:

- The Constitution was amended to prevent elected MLAs and MPs from changing parties. This was done because many elected representatives were indulging in DEFECTION in order to become ministers or for cash rewards. Now the law says that if

Glossary

Defection: Changing party allegiance from the party on which a person got elected (to a legislative body) to a different party.

Reprint 2026-27



© Keshav - *The Hindu*

Do you agree that this form of reforming political parties will be acceptable to them?

any MLA or MP changes parties, he or she will lose the seat in the legislature. This new law has helped bring defection down. At the same time, this has made any dissent even more difficult. MPs and MLAs have to accept whatever the party leaders decide.

- The Supreme Court passed an order to reduce the influence of money and criminals. Now, it is mandatory for every candidate who contests elections to file an AFFIDAVIT giving details of his property and criminal cases pending against him. The new system has made a lot of information available to the public. But there is no system to check if the information given by the candidates is true. As yet we do not know if it has led to decline in the influence of the rich and the criminals.
- The Election Commission passed an order making it necessary for political parties to hold their organisational elections and file their income tax returns. The parties have started doing so but sometimes it is mere formality. It is not clear if this step has led to greater internal democracy in political parties.

Besides these, many suggestions are often made to reform political parties:

- A law should be made to regulate the internal affairs of political parties. It should be made compulsory for political parties to maintain a register of its members, to follow its own constitution, to have an independent authority, to act as a judge in case of party disputes, to hold open elections to the highest posts.
- It should be made mandatory for political parties to give a minimum number of tickets, about one-third, to women candidates. Similarly, there should be a quota for women in the decision making bodies of the party.
- There should be state funding of elections. The government should give parties money to support their election expenses. This support could be given in kind: petrol, paper, telephone, etc. Or it could be given in cash on the basis of the votes secured by the party in the last election.

These suggestions have not yet been accepted by political parties. If and when these are accepted these could lead to some improvement. But we must be very careful about legal solutions to political problems. Over-regulation of political parties can be counter-productive. This would force all parties to find ways to cheat the law. Besides, political parties will not agree to pass a law that they do not like.

There are two other ways in which political parties can be reformed. One, people can put

Glossary

Affidavit: A signed document submitted to an officer, where a person makes a sworn statement regarding her personal information.

Reprint 2026-27

pressure on political parties. This can be done through petitions, publicity and agitations. Ordinary citizens, pressure groups and movements and the media can play an important role in this. If political parties feel that they would lose public support by not taking up reforms, they would become more serious about reforms. Two, political parties can improve if those who want this, join political parties. The quality of democracy depends on the degree of public participation. It is difficult to reform politics if ordinary citizens do not take part in it and simply criticise it from the outside. The problem of bad politics can be solved by more and better politics. We shall return to this theme in the final chapter.

Exercises

1. State the various functions political parties perform in a democracy.
2. What are the various challenges faced by political parties?
3. Suggest some reforms to strengthen parties so that they perform their functions well?
4. What is a political party?
5. What are the characteristics of a political party?
6. A group of people who come together to contest elections and hold power in the government is called a _____.
7. Match List I (organisations and struggles) with List II and select the correct answer using the codes given below the lists:

	List I	List II
1.	Congress Party	A. National Democratic Alliance
2.	Bharatiya Janata Party	B. State party
3.	Communist Party of India (Marxist)	C. United Progressive Alliance
4.	Telugu Desam Party	D. Left Front

	1	2	3	4
(a)	C	A	B	D
(b)	C	D	A	B
(c)	C	A	D	B
(d)	D	C	A	B

8. Who among the following is the founder of the Bahujan Samaj Party?
   A. Kanshi Ram
   B. Sahu Maharaj
   C. B.R. Ambedkar
   D. Jotiba Phule
9. What is the guiding philosophy of the Bharatiya Janata Party?
   A. Bahujan Samaj
   B. Revolutionary democracy
   C. Integral humanism
   D. Modernity



© NCERT not to be republished

Schwann (1839), a German Zoologist, studied different types of animal cells and reported that cells had a thin outer layer which is today known as the 'plasma membrane'. He also concluded, based on his studies on plant tissues, that the presence of cell wall is a unique character of the plant cells. On the basis of this, Schwann proposed the hypothesis that the bodies of animals and plants are composed of cells and products of cells.

Schleiden and Schwann together formulated the cell theory. This theory however, did not explain as to how new cells were formed. Rudolf Virchow (1855) first explained that cells divided and new cells are formed from pre-existing cells (*Omnis cellula-e cellula*). He modified the hypothesis of Schleiden and Schwann to give the cell theory a final shape. Cell theory as understood today is:

(i) all living organisms are composed of cells and products of cells.

(ii) all cells arise from pre-existing cells.

8.3 AN OVERVIEW OF CELL

You have earlier observed cells in an onion peel and/or human cheek cells under the microscope. Let us recollect their structure. The onion cell which is a typical plant cell, has a distinct cell wall as its outer boundary and just within it is the cell membrane. The cells of the human cheek have an outer membrane as the delimiting structure of the cell. Inside each cell is a dense membrane bound structure called nucleus. This nucleus contains the chromosomes which in turn contain the genetic material, DNA. Cells that have membrane bound nuclei are called eukaryotic whereas cells that lack a membrane bound nucleus are prokaryotic. In both prokaryotic and eukaryotic cells, a semi-fluid matrix called cytoplasm occupies the volume of the cell. The cytoplasm is the main arena of cellular activities in both the plant and animal cells. Various chemical reactions occur in it to keep the cell in the 'living state'.

Besides the nucleus, the eukaryotic cells have other membrane bound distinct structures called **organelles** like the endoplasmic reticulum (ER), the golgi complex, lysosomes, mitochondria, microbodies and vacuoles. The prokaryotic cells lack such membrane bound organelles.

Ribosomes are non-membrane bound organelles found in all cells – both eukaryotic as well as prokaryotic. Within the cell, ribosomes are found not only in the cytoplasm but also within the two organelles – chloroplasts (in plants) and mitochondria and on rough ER.

Animal cells contain another non-membrane bound organelle called centrosome which helps in cell division.

Cells differ greatly in size, shape and activities (Figure 8.1). For example, Mycoplasmas, the smallest cells, are only 0.3 μm in length while bacteria

© NCERT not to be republished

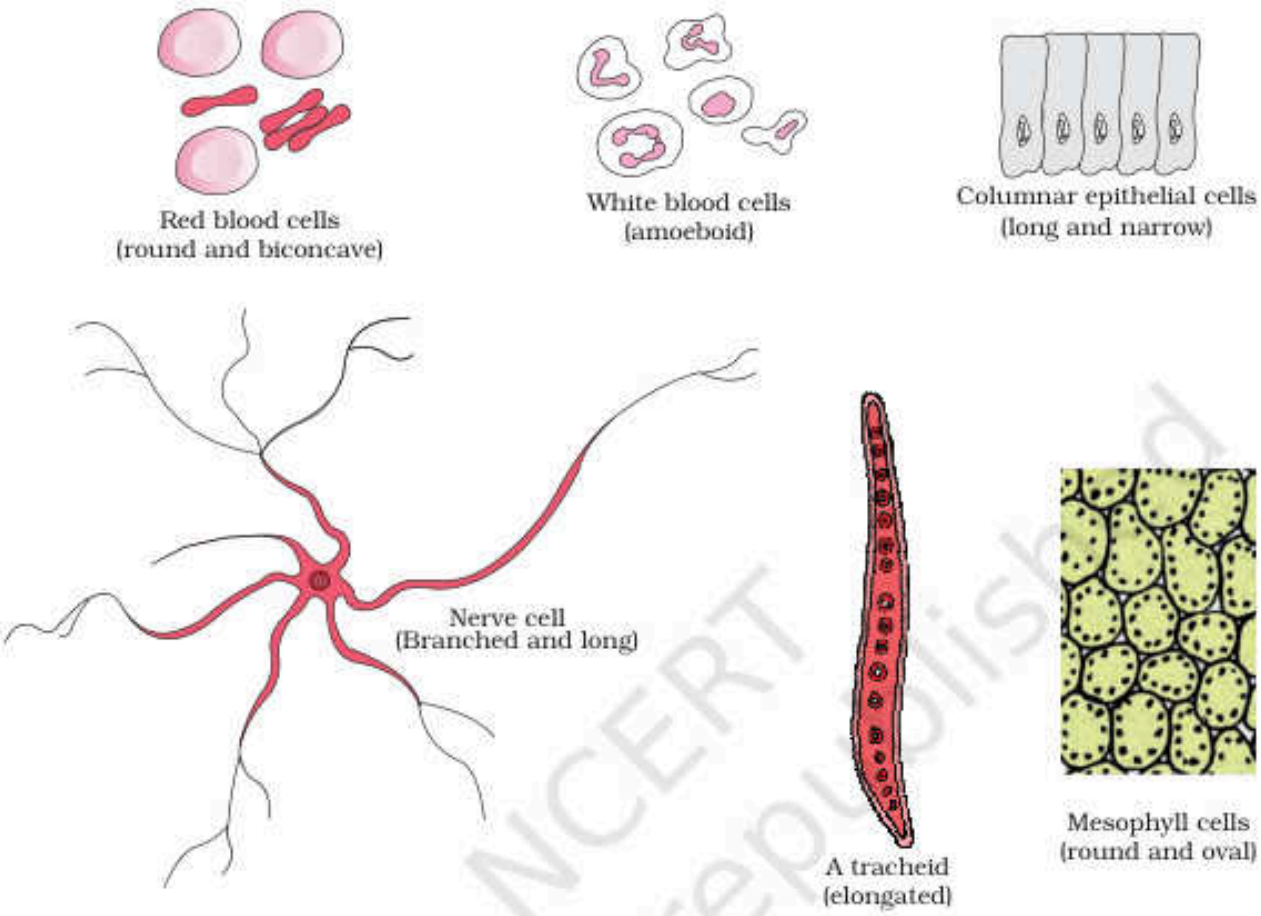


Figure 8.1 Diagram showing different shapes of the cells

could be 3 to 5 µm. The largest isolated single cell is the egg of an ostrich. Among multicellular organisms, human red blood cells are about 7.0 µm in diameter. Nerve cells are some of the longest cells. Cells also vary greatly in their shape. They may be disc-like, polygonal, columnar, cuboid, thread like, or even irregular. The shape of the cell may vary with the function they perform.

8.4 PROKARYOTIC CELLS

The prokaryotic cells are represented by bacteria, blue-green algae, mycoplasma and PPLO (Pleuro Pneumonia Like Organisms). They are generally smaller and multiply more rapidly than the eukaryotic cells (Figure 8.2). They may vary greatly in shape and size. The four basic shapes of bacteria are bacillus (rod like), coccus (spherical), vibrio (comma shaped) and spirillum (spiral).

The organisation of the prokaryotic cell is fundamentally similar even though prokaryotes exhibit a wide variety of shapes and functions. All

© NCERT not to be republished

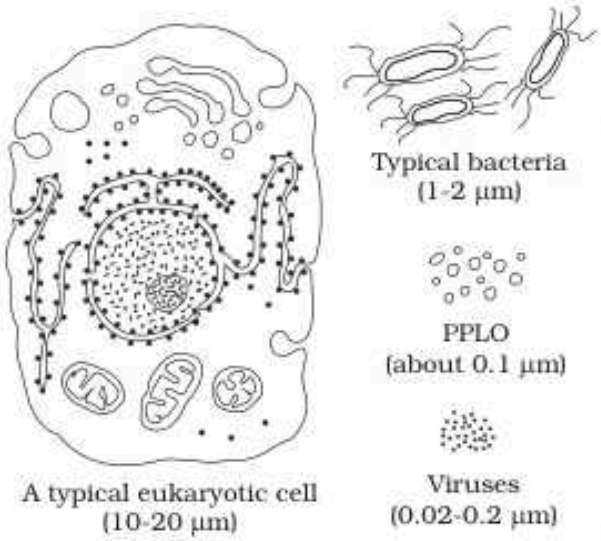


Figure 8.2 Diagram showing comparison of eukaryotic cell with other organisms

prokaryotes have a cell wall surrounding the cell membrane except in mycoplasma. The semi-fluid matrix filling the cell is the cytoplasm. There is no well-defined nucleus. The genetic material is basically naked, not enveloped by a nuclear membrane. In addition to the genomic DNA (the single chromosome/circular DNA), many bacteria have small circular DNA outside the genomic DNA. These smaller DNA are called plasmids. The plasmid DNA confers certain unique phenotypic characters to such bacteria. One such character is resistance to antibiotics. In higher classes you will learn that this plasmid DNA is used to monitor bacterial transformation with foreign DNA. Nuclear membrane is found in eukaryotes. No organelles, like the ones in eukaryotes, are found in prokaryotic cells except for ribosomes. Prokaryotes have something unique in the form of inclusions. A specialised differentiated form of cell membrane called mesosome is the characteristic of prokaryotes. They are essentially infoldings of cell membrane.

8.4.1 Cell Envelope and its Modifications

Most prokaryotic cells, particularly the bacterial cells, have a chemically complex cell envelope. The cell envelope consists of a tightly bound three layered structure i.e., the outermost glycocalyx followed by the cell wall and then the plasma membrane. Although each layer of the envelope performs distinct function, they act together as a single protective unit. Bacteria can be classified into two groups on the basis of the differences in the cell envelopes and the manner in which they respond to the staining procedure developed by Gram viz., those that take up the gram stain are **Gram positive** and the others that do not are called **Gram negative** bacteria.

Glycocalyx differs in composition and thickness among different bacteria. It could be a loose sheath called the **slime layer** in some, while in others it may be thick and tough, called the **capsule**. The **cell wall** determines the shape of the cell and provides a strong structural support to prevent the bacterium from bursting or collapsing.

The plasma membrane is selectively permeable in nature and interacts with the outside world. This membrane is similar structurally to that of the eukaryotes.

A special membranous structure is the **mesosome** which is formed by the extensions of plasma membrane into the cell. These extensions are in the **form of vesicles, tubules and lamellae**. They help in cell wall

© NCERT not to be republished

formation, DNA replication and distribution to daughter cells. They also help in respiration, secretion processes, to increase the surface area of the plasma membrane and enzymatic content. In some prokaryotes like cyanobacteria, there are other membranous extensions into the cytoplasm called chromatophores which contain pigments.

Bacterial cells may be motile or non-motile. If motile, they have thin filamentous extensions from their cell wall called flagella. Bacteria show a range in the number and arrangement of flagella. Bacterial flagellum is composed of three parts – **filament, hook** and **basal body**. The filament is the longest portion and extends from the cell surface to the outside.

Besides flagella, Pili and Fimbriae are also surface structures of the bacteria but do not play a role in motility. The **pili** are elongated tubular structures made of a special protein. The **fimbriae** are small bristle like fibres sprouting out of the cell. In some bacteria, they are known to help attach the bacteria to rocks in streams and also to the host tissues.

8.4.2 Ribosomes and Inclusion Bodies

In prokaryotes, ribosomes are associated with the plasma membrane of the cell. They are about 15 nm by 20 nm in size and are made of two subunits - 50S and 30S units which when present together form 70S prokaryotic ribosomes. Ribosomes are the site of protein synthesis. Several ribosomes may attach to a single mRNA and form a chain called **polyribosomes** or **polysome**. The ribosomes of a polysome translate the mRNA into proteins.

Inclusion bodies: Reserve material in prokaryotic cells are stored in the cytoplasm in the form of inclusion bodies. These are not bound by any membrane system and lie free in the cytoplasm, e.g., phosphate granules, cyanophycean granules and glycogen granules. Gas vacuoles are found in blue green and purple and green photosynthetic bacteria.

8.5 EUKARYOTIC CELLS

The eukaryotes include all the protists, plants, animals and fungi. In eukaryotic cells there is an extensive compartmentalisation of cytoplasm through the presence of membrane bound organelles. Eukaryotic cells possess an organised nucleus with a nuclear envelope. In addition, eukaryotic cells have a variety of complex locomotory and cytoskeletal structures. Their genetic material is organised into chromosomes.

All eukaryotic cells are not identical. Plant and animal cells are different as the former possess cell walls, plastids and a large central vacuole which are absent in animal cells. On the other hand, animal cells have centrioles which are absent in almost all plant cells (Figure 8.3).

© NCERT not to be republished

The central objective of Planning in India... is to initiate a process of development which will raise the living standards and open out to the people new opportunities for a richer and more varied life.

First Five Year Plan

2.1 Introduction

On 15 August 1947, India woke to a new dawn of freedom. Finally we were masters of our own destiny after some two hundred years of British rule; the job of nation building was now in our own hands. The leaders of independent India had to decide, among other things, the type of **economic system** most suitable for our nation, a system which would promote the welfare of all rather than a few. There are different types of economic systems (see Box 2.1) and among them, socialism appealed to Jawaharlal Nehru the most. However, he was not in favour of the kind of socialism established in the former Soviet Union where all the means of production, i.e. all the factories and farms in the country, were owned by the government. There was no private property. It is not possible in a democracy like India for the government to change the ownership pattern of land and other properties of its citizens in the way that it was done in the former Soviet Union.

Nehru, and many other leaders and thinkers of the newly independent India, sought an alternative to the extreme versions of capitalism and socialism. Basically sympathising with the socialist outlook, they found the answer in an economic system which, in their view, combined the best features of socialism without its drawbacks. In this view, India would be a socialist society with a strong public sector but also with private property and democracy; the government would plan (see Box 2.2) for the



Work These Out

- Prepare a chart on the different types of economic systems prevalent in the world. List out the countries as capitalist, socialist and mixed economy.
- Plan a class trip to an agriculture farm. Divide the class into seven groups with each group to plan a specific goal, for example, the purpose of the visit, money expenditure involved, time taken, resources, people accompanying the group and who need to be contacted, possible places of visit, possible questions to be asked etc. Now, with the help of your teacher, compile these specific goals and compare with long-term goals of successful visit to an agricultural farm.

Reprint 2026-27

Box 2.1: Types of Economic Systems

Every society has to answer three questions

- What goods and services should be produced in the country?
- How should the goods and services be produced? Should producers use more human labour or more capital (machines) for producing things?
- How should the goods and services be distributed among people?

One answer to these questions is to depend on the **market forces** of supply and demand. In a market economy, also called capitalism, only those consumer goods will be produced that are in demand, i.e., goods that can be sold profitably either in the domestic or in the foreign markets. If cars are in demand, cars will be produced and if bicycles are in demand, bicycles will be produced. If labour is cheaper than capital, more labour-intensive methods of production will be used and vice-versa. In a **capitalist** society the goods produced are distributed among people not on the basis of what people need but on the basis of Purchasing Power—the ability to buy goods and services. That is, one has to have the money in the pocket to buy it. Low cost housing for the poor is much needed but will not count as demand in the market sense because the poor do not have the purchasing power to back the demand. As a result this commodity will not be produced and supplied as per market forces. Such a society did not appeal to Jawaharlal Nehru, our first prime minister, for it meant that the great majority of people of the country would be left behind without the chance to improve their quality of life.

A **socialist** society answers the three questions in a totally different manner. In a socialist society the government decides what goods are to be produced in accordance with the needs of society. It is assumed that the government knows what is good for the people of the country and so the desires of individual consumers are not given much importance. The government decides how goods are to be produced and how they should be distributed. In principle, distribution under socialism is supposed to be based on what people need and not on what they can afford to purchase. Unlike under capitalism, for example, a socialist nation provides free health care to all its citizens. Strictly, a socialist society has no private property since everything is owned by the state. In Cuba and China, for example, most of the economic activities are governed by the socialistic principles.

Most economies are **mixed economies,** i.e. the government and the market together answer the three questions of what to produce, how to produce and how to distribute what is produced. In a mixed economy, the market will provide whatever goods and services it can produce well, and the government will provide essential goods and services which the market fails to do.

Reprint 2026-27

Box 2.2: What is a Plan?

A plan spells out how the resources of a nation should be put to use. It should have some general goals as well as specific objectives which are to be achieved within a specified period of time; in India plans were of five years duration and were called five year plans (we borrowed this from the former Soviet Union, the pioneer in national planning). Our plan documents upto the year 2017 not only specify the objectives to be attained in the five years of a plan but also what is to be achieved over a period of twenty years. This long-term plan is called 'perspective plan'. The five year plans were supposed to provide the basis for the perspective plan.

It will be unrealistic to expect all the goals of a plan to be given equal importance in all the plans. In fact the goals may actually be in conflict. For example, the goal of introducing modern technology may be in conflict with the goal of increasing employment if the technology reduces the need for labour. The planners have to balance the goals, a very difficult job indeed. We find different goals being emphasised in different plans in India.

India's five year plans did not spell out how much of each and every good and service is to be produced. This is neither possible nor necessary (the former Soviet Union tried to do this and failed). It is enough if the plan is specific about the sectors where it plays a commanding role, for instance, power generation and irrigation, while leaving the rest to the market.

economy with the private sector being encouraged to be part of the plan effort. The 'Industrial Policy Resolution' of 1948 and the Directive Principles of the Indian Constitution reflected this outlook. In 1950, the **Planning Commission** was set up with the Prime Minister as its Chairperson. The era of five year plans had begun.

2.2 The Goals of Five Year Plans

A plan should have some clearly specified goals. The goals of the five year plans were: growth, modernisation, self-reliance and equity. This does not mean that all the plans have given equal importance to all these goals. Due to limited resources, a choice has to be made in each plan about which of the goals is to be given primary importance. Nevertheless, the planners have to ensure that, as far as possible, the policies of the plans do not contradict these four goals. Let us now learn about the goals of planning in some detail.

Growth: It refers to increase in the country's capacity to produce the output of goods and services within the country. It implies either a larger stock of productive capital, or a larger size of supporting services like transport and banking, or an increase in the efficiency of productive capital and services. A good indicator of economic growth, in the language of

Reprint 2026-27

Box 2.3: Mahalanobis: the Architect of Indian Planning

Many distinguished thinkers contributed to the formulation of India's five year plans. Among them, the name of the statistician, Prasanta Chandra Mahalanobis, stands out.

Planning, in the real sense of the term, began with the Second Five Year Plan. The Second Plan, a landmark contribution to development planning in general, laid down the basic ideas regarding goals of Indian planning; this plan was based on the ideas of Mahalanobis. In that sense, he can be regarded as the architect of Indian planning.



Mahalanobis was born in 1893 in Calcutta. He was educated at the Presidency College in Calcutta and at Cambridge University in England. His contributions to the subject of statistics brought him international fame. In 1945 he was made a Fellow (member) of Britain's Royal Society, one of the most prestigious organisations of scientists; only the most outstanding scientists are made members of this Society.

Mahalanobis established the Indian Statistical Institute (ISI) in Calcutta and started a journal, *Sankhya*, which still serves as a respected forum for statisticians to discuss their ideas. Both, the ISI and *Sankhya*, are highly regarded by statisticians and economists all over the world to this day.

During the second plan period, Mahalanobis invited many distinguished economists from India and abroad to advise him on India's economic development. Some of these economists became Nobel Prize winners later, which shows that he could identify individuals with talent. Among the economists invited by Mahalanobis were those who were very critical of the socialist principles of the second plan. In other words, he was willing to listen to what his critics had to say, the mark of a great scholar.

Many economists today reject the approach to planning formulated by Mahalanobis but he will always be remembered for playing a vital role in putting India on the road to economic progress, and statisticians continue to profit from his contribution to statistical theory.

Source: *Sukhamoy Chakravarty, 'Mahalanobis, Prasanta Chandra' in John Eatwell et.al, (Eds.) The New Palgrave Dictionary: Economic Development, W.W. Norton, New York and London.*

Reprint 2026-27

Box 2.4: The Service Sector

As a country develops, it undergoes 'structural change'. In the case of India, the structural change is peculiar. Usually, with development, the share of agriculture declines and the share of industry becomes dominant. At higher levels of development, the service sector contributes more to the GDP than the other two sectors. In India, the share of agriculture in the GDP was more than 50 per cent—as we would expect for a poor country. But by 1990 the share of the service sector was 40.59 per cent, more than that of agriculture or industry, like what we find in developed nations. This phenomenon of growing share of the service sector was accelerated in the post 1991 period (this marked the onset of globalisation in the country which will be discussed in chapter 3).

economics, is steady increase in the Gross Domestic Product (GDP). The GDP is the market value of all the final goods and services produced in the country during a year. You have studied this concept in Class X as well. You can think of the GDP as a cake and growth is increase in the size of the cake. If the cake is larger, more people can enjoy it. It is necessary to produce more goods and services if the people of India are to enjoy (in the words of the First Five Year Plan) a more rich and varied life.

The GDP of a country is derived from the different sectors of the economy, namely the agricultural sector, the industrial sector and the service sector. The contribution made by each of these sectors makes up the **structural composition** of the economy. In some countries, growth in agriculture contributes more to the GDP growth, while in some countries the growth in the service sector contributes more to GDP growth (see Box 2.4).

Modernisation: To increase the production of goods and services the producers have to adopt new technology. For example, a farmer can increase the output on the farm by using new seed varieties instead of using the old ones. Similarly, a factory can increase output by using a new type of machine. Adoption of new technology is called modernisation.

However, modernisation does not refer only to the use of new technology but also to changes in social outlook such as the recognition that women should have the same rights as men. In a traditional society, women are supposed to remain at home while men work. A modern society makes use of the talents of women in the work place — in banks, factories, schools etc. — and such a society in most occassions is also prosperous.

Self-reliance: A nation can promote economic growth and modernisation by using its own resources or by using resources imported from other nations. The first seven five year plans gave importance to self-reliance which means avoiding imports of those goods which could be

Reprint 2026-27



Work These Out

- Discuss in your class the changes in technology used for
  (a) Production of food grains
  (b) Packaging of products
  (c) Mass communication
- Find out and prepare a list of major items that India imported and exported during 1990-91 and 2018-19. (For this, see P. 145 also).
  (a) Observe the difference
  (b) Do you see the impact of self-reliance? Discuss.

For getting these details you may refer to *Economic Survey* of the latest year.

produced in India itself. This policy was considered a necessity in order to reduce our dependence on foreign countries, especially for food. It is understandable that people who were recently freed from foreign domination should give importance to self-reliance. Further, it was feared that dependence on imported food supplies, foreign technology and foreign capital may make India's sovereignty vulnerable to foreign interference in our policies.

Equity: Now growth, modernisation and self-reliance, by themselves, may not improve the kind of life which people are living. A country can have high growth, the most modern technology developed in the country itself, and also have most of its people living in poverty. It is important to ensure that the benefits of economic prosperity reach the poor sections as well instead of being enjoyed only by the rich. So, in addition to growth, modernisation and self-reliance, equity is also important. Every Indian should be able to meet his or her basic needs such as food, a decent house, education and health care and inequality in the distribution of wealth should be reduced.

Let us now see how the first seven five year plans, covering the period 1950-1990, attempted to attain these four goals and the extent to which they succeeded in doing so, with reference to agriculture, industry and trade. You will study the policies and developmental issues taken up after 1991 in Chapter 3.

2.3 Agriculture

You have learnt in Chapter 1 that during the colonial rule there was neither growth nor equity in the agricultural sector. The policy makers of independent India had to address these issues which they did through land reforms and promoting the use of 'High Yielding Variety' (HYV) seeds which ushered in a revolution in Indian agriculture.

Land Reforms: At the time of independence, the land tenure system was characterised by intermediaries

Reprint 2026-27

Box 2.5: Ownership and Incentives

The policy of 'land to the tiller' is based on the idea that the cultivators will take more interest—they will have more incentive—in increasing output if they are the owners of the land. This is because ownership of land enables the tiller to make profit from the increased output. Tenants do not have the incentive to make improvements on land since it is the landowner who would benefit more from higher output. The importance of ownership in providing incentives is well illustrated by the carelessness with which farmers in the former Soviet Union used to pack fruits for sale. It was not uncommon to see farmers packing rotten fruits along with fresh fruits in the same box. Now, every farmer knows that the rotten fruits will spoil the fresh fruits if they are packed together. This will be a loss to the farmer since the fruits cannot be sold. So why did the Soviet farmers do something which would so obviously result in loss for them? The answer lies in the incentives facing the farmers. Since farmers in the former Soviet Union did not own any land, they neither enjoyed the profits nor suffered the losses. In the absence of ownership, there was no incentive on the part of farmers to be efficient, which also explains the poor performance of the agricultural sector in the Soviet Union despite availability of vast areas of highly fertile land.

Source: *Thomas Sowell, Basic Economics: A Citizen's Guide to the Economy, New York: Basic Books, 2004, Second Edition.*

(variously called *zamindars, jagirdars* etc.) who merely collected rent from the actual tillers of the soil without contributing towards improvements on the farm. The low productivity of the agricultural sector forced India to import food from the United States of America (U.S.A.). Equity in agriculture called for **land reforms** which primarily refer to change in the ownership of landholdings. Just a year after independence, steps were taken to abolish intermediaries and to make the tillers the owners of land. The idea behind this move was that ownership of land would give incentives (see Box 2.5) to the tillers to invest in making improvements provided sufficient capital was made available to them.

Land ceiling was another policy to promote equity in the agricultural sector. This means fixing the maximum size of land which could be owned by an individual. The purpose of land ceiling was to reduce the concentration of land ownership in a few hands.

The abolition of intermediaries meant that some 200 lakh tenants came into direct contact with the government — they were thus freed from being exploited by the *zamindars*. The ownership conferred on tenants gave them the incentive to increase output and this contributed to growth in agriculture. However, the goal of equity was not fully served by abolition of intermediaries. In some areas the former *zamindars*

Reprint 2026-27

continued to own large areas of land by making use of some loopholes in the legislation; there were cases where tenants were evicted and the landowners claimed to be self-cultivators (the actual tillers), claiming ownership of the land; and even when the tillers got ownership of land, the poorest of the agricultural labourers (such as sharecroppers and landless labourers) did not benefit from land reforms.

The land ceiling legislation also faced hurdles. The big landlords challenged the legislation in the courts, delaying its implementation. They used this delay to register their lands in the name of close relatives, thereby escaping from the legislation. The legislation also had a lot of loopholes which were exploited by the big landholders to retain their land. Land reforms were successful in Kerala and West Bengal because these states had governments committed to the policy of land to the tiller. Unfortunately other states did not have the same level of commitment and vast inequality in landholding continues to this day.

The Green Revolution: At independence, about 75 per cent of the country's population was dependent on agriculture. Productivity in the agricultural sector was very low because of the use of old technology and the absence of required infrastructure for the vast majority of farmers. India's agriculture vitally depends on the monsoon and if the monsoon fell short the farmers were in trouble unless they had access to irrigation facilities which very few had. The stagnation in agriculture during the colonial rule was permanently broken by the green revolution. This refers to the large increase in production of food grains resulting from the use of high yielding variety (HYV) seeds especially for wheat and rice. The use of these seeds required the use of fertiliser and pesticide in the correct quantities as well as regular supply of water; the application of these inputs in correct proportions is vital. The farmers who could benefit from HYV seeds required reliable irrigation facilities as well as the financial resources to purchase fertiliser and pesticide. As a result, in the first phase of the green revolution (approximately mid 1960s upto mid 1970s), the use of HYV seeds was restricted to the more affluent states such as Punjab, Andhra Pradesh and Tamil Nadu. Further, the use of HYV seeds primarily benefited the wheat-growing regions only. In the second phase of the green revolution (mid-1970s to mid-1980s), the HYV technology spread to a larger number of states and benefited more variety of crops. The spread of green revolution technology enabled India to achieve self-sufficiency in food grains; India no longer had to be at the mercy of America, or any other nation, for meeting its food requirements.

Growth in agricultural output is important but it is not enough. If a large proportion of this increase is

Reprint 2026-27



consumed by the farmers themselves instead of being sold in the market, the higher output will not make much of a difference to the economy as a whole. If, on the other hand, a substantial amount of agricultural produce is sold in the market by the farmers, the higher output can make a difference to the economy. The portion of agricultural produce which is sold in the market by the farmers is called **marketed surplus**. A good proportion of the rice and wheat produced during the green revolution period (available as marketed surplus) was sold by the farmers in the market. As a result, the price of food grains declined relative to other items of consumption. The low-income groups, who spend a large percentage of their income on food, benefited from this decline in relative prices. The green revolution enabled the government to procure sufficient amount of food grains to build a stock which could be used in times of food shortage.

While the nation had immensely benefited from the green revolution, the technology involved was not free from risks. One such risk was the possibility that it would increase the disparities between small and big farmers—since only the big farmers could afford the required inputs, thereby reaping most of the benefits of the green revolution. Moreover, the HYV crops were also more prone to attack by pests and the small farmers who adopted this technology could lose everything in a pest attack.

Reprint 2026-27

Fortunately, these fears did not come true because of the steps taken by the government. The government provided loans at a low interest rate to small farmers and subsidised fertilisers so that small farmers could also have access to the needed inputs. Since the small farmers could obtain the required inputs, the output on small farms equalled the output on large farms in the course of time. As a result, the green revolution benefited the small as well as rich farmers. The risk of the small farmers being ruined when pests attack their crops was considerably reduced by the services rendered by research institutes established by the government. You should note that the green revolution would have favoured the rich farmers only if the state did not play an extensive role in ensuring that the small farmer also gains from the new technology.

The Debate Over Subsidies: The economic justification of subsidies in agriculture is, at present, a hotly debated question. It is generally agreed that it was necessary to use subsidies to provide an incentive for adoption of the new HYV technology by farmers in general and small farmers in particular. Any new technology will be looked upon as being risky by farmers. Subsidies were, therefore, needed to encourage farmers to test the new technology. Some economists believe that once the technology is found profitable and is widely adopted, subsidies should be phased out since their purpose has been served. Further, subsidies are meant to benefit the farmers but a substantial amount of fertiliser subsidy also benefits the fertiliser industry; and among farmers, the subsidy largely benefits the farmers in the more prosperous regions. Therefore, it is argued that there is no case for continuing with fertiliser subsidies; it does not benefit the target group and it is a huge burden on the government's finances (see also Box 2.6).

On the other hand, some believe that the government should continue with agricultural subsidies because farming in India continues to be a risky business. Most farmers are very poor and they will not be able to afford the required inputs without subsidies. Eliminating subsidies will increase the inequality between rich and poor farmers and violate the goal of equity. These experts argue that if subsidies are largely benefiting the fertiliser industry and big farmers, the correct policy is not to abolish subsidies but to take steps to ensure that only the poor farmers enjoy the benefits.

Thus, by the late 1960s, Indian agricultural productivity had increased sufficiently to enable the country to be self-sufficient in food grains. This is an achievement to be proud of. On the negative side, some 65 per cent of the country's population continued to be employed in agriculture even as late as 1990. Economists have found that as

Reprint 2026-27

Box 2.6: Prices as Signals

You would have learnt in an earlier class about how prices of goods are determined in the market. It is important to understand that prices are signals about the availability of goods. If a good becomes scarce, its price will rise and those who use this good will have the incentive to make efficient decisions about its use based on the price. If the price of water goes up because of lower supply, people will have the incentive to use it with greater care; for example, they may stop watering the garden to conserve water. We complain whenever the price of petrol increases and blame it on the government. But the increase in petrol price reflects greater scarcity and the price rise is a signal that less petrol is available—this provides an incentive to use less petrol or look for alternate fuels.

Some economists point out that subsidies do not allow prices to indicate the supply of a good. When electricity and water are provided at a subsidised rate or free, they will be used wastefully without any concern for their scarcity. Farmers will cultivate water intensive crops if water is supplied free, although the water resources in that region may be scarce and such crops will further deplete the already scarce resources. If water is priced to reflect scarcity, farmers will cultivate crops suitable to the region. Fertiliser and pesticide subsidies result in overuse of resources which can be harmful to the environment. Subsidies provide an incentive for wasteful use of resources. Think about subsidies in terms of incentives and ask yourself whether it is wise from the economic viewpoint to provide free electricity to farmers.

a nation becomes more prosperous, the proportion of GDP contributed by agriculture as well as the proportion of population working in the sector declines considerably. In India, between 1950 and 1990, the proportion of GDP contributed by agriculture declined significantly but not the population depending on it (67.5 per cent in 1950 to 64.9 per cent by 1990). Why was such a large proportion of the population engaged in agriculture although agricultural output could have grown with much less people working in the sector? The answer is that the industrial sector and the service sector did not absorb the people working in the agricultural sector. Many economists call this an important failure of our policies followed during 1950-1990.

2.4 Industry and Trade

Economists have found that poor nations can progress only if they have a good industrial sector. Industry provides employment which is more stable than the employment in agriculture; it promotes modernisation and overall prosperity. It is for this reason that the five year plans placed a lot of emphasis on industrial development. You might have studied in the previous chapter that, at the time of independence, the variety of industries was very narrow

Reprint 2026-27

investment in industrial ventures required for the development of Indian economy; nor was the market big enough to encourage industrialists to undertake major projects even if they had the capital to do so. It is principally for these reasons that the erstwhile governments had to play an extensive role in promoting the industrial sector. In addition, the decision to develop the Indian economy on socialist lines led to the policy of the government controlling the commanding heights of the economy, as the Second Five Year plan put it. This meant that the government would have complete control of those industries that were vital for the economy. The policies of the private sector would have to be complimentary to those of the public sector, with the public sector leading the way.

Industrial Policy Resolution 1956 (IPR 1956): In accordance with the goal of the state controlling the commanding heights of the economy, the Industrial Policy Resolution of 1956 was adopted. This resolution formed the basis of the Second Five Year Plan, the plan which tried to build the basis for a socialist pattern of society. This resolution classified industries into three categories. The first category comprised industries which would be exclusively owned by the government; the second category consisted of industries in which the private sector could supplement the efforts of the public sector, with the government taking the sole responsibility for starting new units; the third category consisted of the remaining industries which were to be in the private sector.

Although there was a category of industries left to the private sector, the sector was kept under state control through a system of licenses. No new industry was allowed unless a license was obtained from the government. This policy was used for promoting industry in backward regions; it was easier to obtain a license if the industrial unit was established in an economically backward area. In addition, such units were given certain concessions such as tax benefits and electricity at a lower tariff. The purpose of this policy was to promote **regional equality**.

Even an existing industry had to obtain a license for expanding output or for diversifying production (producing a new variety of goods). This was meant to ensure that the quantity of goods produced was not more than what the economy required. License to expand production was given only if the government was convinced that the economy required a larger quantity of goods.

Small-Scale Industry: In 1955, the Village and Small-Scale Industries Committee, also called the Karve Committee, noted the possibility of using small-scale industries for promoting rural development. A 'small-scale industry' is defined with reference to the maximum investment allowed on the assets of a unit. This limit has changed over a period of time. In 1950 a small-scale industrial unit was one which invested

Reprint 2026-27

a maximum of rupees five lakh; at present the maximum investment allowed is rupees one crore.

It is believed that small-scale industries are more 'labour intensive' i.e., they use more labour than the large-scale industries and, therefore, generate more employment. But these industries cannot compete with the big industrial firms; it is obvious that development of small-scale industry requires them to be shielded from the large firms. For this purpose, the production of a number of products was reserved for the small-scale industry; the criterion of reservation being the ability of these units to manufacture the goods. They were also given concessions such as lower excise duty and bank loans at lower interest rates.

2.5 Trade Policy: Import Substitution

The industrial policy that India adopted was closely related to the **trade policy**. In the first seven plans, trade was characterised by what is commonly called an inward looking trade strategy. Technically, this strategy is called **import substitution**. This policy aimed at replacing or substituting imports with domestic production. For example, instead of importing vehicles made in a foreign country, industries would be encouraged to produce them in India itself. In this policy the government protected the domestic industries from foreign competition. Protection from imports took two forms: tariffs and quotas. Tariffs are a tax on imported goods; they make imported goods more expensive and discourage their use. Quotas specify the quantity of goods which can be imported. The effect of tariffs and quotas is that they restrict imports and, therefore, protect the domestic firms from foreign competition.

The policy of protection was based on the notion that industries of developing countries were not in a position to compete against the goods produced by more developed economies. It was assumed that if the domestic industries were protected they would learn to compete in the course of time. Our planners also feared the possibility of foreign exchange being spent on import of luxury goods if no restrictions were placed on imports. Nor was any serious thought given to promote exports until the mid-1980s.

Effect of Policies on Industrial Development: The achievements of India's industrial sector during the first seven plans are impressive indeed. The proportion of GDP contributed by the industrial sector increased in the period from 13 per cent in 1950-51 to 24.6 per cent in 1990-91. The rise in the industry's share of GDP is an important indicator of development. The six per cent annual growth rate of the industrial sector during the period is commendable. No longer was Indian industry restricted largely to cotton textiles and jute; in fact, the industrial sector became well diversified by 1990, largely due to

Reprint 2026-27

Work These Out

- Construct a pie chart for the following table on sectoral contribution to GDP and discuss the difference in the contribution of the sectors in the light of effects of development during 1950-91.

Sector	*1950-51*	*1990-91*
Agriculture 	59.0	34.9
Industry 	13.0	24.6
Services 	28.0	40.5

- Conduct a debate in your classroom on the usefulness of Public Sector Undertakings (PSUs) by dividing the class into two groups. One group may speak in favour of PSUs and the other group against the motion (involve as many students as possible and encourage them to give examples).

the public sector. The promotion of small-scale industries gave opportunities to those people who did not have the capital to start large firms to get into business. Protection from foreign competition enabled the development of indigenous industries in the areas of electronics and automobile sectors which otherwise could not have developed.

In spite of the contribution made by the public sector to the growth of the Indian economy, some economists are critical of the performance of many public sector enterprises. It was proposed at the beginning of this chapter that initially public sector was required in a big way. It is now widely held that state enterprises continued to produce certain goods and services (often monopolising them) although this was no longer required. An example is the provision of telecommunication service. This industry continued to be reserved for the Public Sector even after it was realised that private sector firms could also provide it. Due to the absence of competition, even till the late 1990s, one had to wait for a long time to get a telephone connection. Another instance could be the establishment of Modern Bread, a bread-manufacturing firm, as if the

Reprint 2026-27

private sector could not manufacture bread! In 2001 this firm was sold to the private sector. The point is that after four decades of Planned development of Indian Economy no distinction was made between (i) what the public sector alone can do and (ii) what the private sector can also do. For example, even now only the public sector supplies national defense. And even though the private sector can manage hotels well, yet, the government also runs hotels. This has led some scholars to argue that the state should get out of areas which the private sector can manage and the government may concentrate its resources on important services which the private sector cannot provide.

Many public sector firms incurred huge losses but continued to function because it is difficult to close a government undertaking even if it is a drain on the nation's limited resources. This does not mean that private firms are always profitable (indeed, quite a few of the public sector firms were originally private firms which were on the verge of closure due to losses; they were then nationalised to protect the jobs of the workers). However, a loss-making private firm will not waste resources by being kept running despite the losses.

The need to obtain a license to start an industry was misused by industrial houses; a big industrialist would get a license not for starting a new firm but to prevent competitors from starting new firms. The excessive regulation of what came to be called the **permit license raj** prevented certain firms from becoming more efficient. More time was spent by industrialists in trying to obtain a license or lobby with the concerned ministries rather than on thinking about how to improve their products.

The protection from foreign competition was also being criticised on the ground that it continued even after it proved to do more harm than good. Due to restrictions on imports, the Indian consumers had to purchase whatever the Indian producers produced. The producers were aware that they had a captive market; so they had no incentive to improve the quality of their goods. Why should they think of improving quality when they could sell low quality items at a high price? Competition from imports forces our producers to be more efficient.

A few economists also point out that the public sector is not meant for earning profits but to promote the welfare of the nation. The public sector firms, on this view, should be evaluated on the basis of the extent to which they contribute to the welfare of people and not on the profits they earn. Regarding protection, some economists hold that we should protect our producers from foreign competition as long as the rich nations continue to do so. Owing to all these conflicts, economists called for a change in our policy. This, alongwith other problems, led the government to introduce a new economic policy in 1991.

Reprint 2026-27

There is a consensus in the world today that economic development is not all and the GDP is not necessarily a measure of progress of a society.

K.R. Narayanan, Former President of India

3.1 Introduction

You have studied in the previous chapter that, since independence, India followed the mixed economy framework by combining the advantages of the capitalist economic system with those of the socialist economic system. Some scholars argue that, over the years, this policy resulted in the establishment of a variety of rules and laws, which were aimed at controlling and regulating the economy, ended up instead in hampering the process of growth and development. Others state that India, which started its developmental path from near stagnation, has since been able to achieve growth in savings, developed a diversified industrial sector which produces a variety of goods and has experienced sustained expansion of agricultural output which has ensured food security.

In 1991, India met with an economic crisis relating to its external debt — the government was not able to make repayments on its borrowings from abroad; **foreign exchange reserves**, which we generally maintain to import petroleum and other important items, dropped to levels that were not sufficient for even a fortnight. The crisis was further compounded by rising prices of essential goods. All these led the government to introduce a new set of policy measures which changed the direction of our developmental strategies. In this chapter, we will look at the background of the crisis, measures that the government has adopted and their impact on various sectors of the economy.

3.2 Background

The origin of the financial crisis can be traced from the inefficient management of the Indian economy in the 1980s. We know that for implementing various policies and its general administration, the government generates funds from various sources such as taxation, running of public sector enterprises etc. When expenditure is more than income, the government borrows to finance the **deficit** from banks and also from people within the country and from international **financial institutions**. When we import goods like petroleum, we pay in dollars which we earn from our exports.

Development policies required that even though the revenues were very low, the government had to overshoot its revenue to meet challenges like unemployment, poverty and population explosion. The continued spending on development programmes of the government did not generate additional revenue. Moreover, the government was not able to

Reprint 2026-27

generate sufficiently from internal sources such as taxation. When the government was spending a large share of its income on areas which do not provide immediate returns such as the social sector and defence, there was a need to utilise the rest of its revenue in a highly efficient manner. The income from public sector undertakings was also not very high to meet the growing expenditure. At times, our **foreign exchange** reserves borrowed from other countries and international financial institutions, was spent on meeting consumption needs. Neither was an attempt made to reduce such profligate spending nor sufficient attention was given to boost exports to pay for the growing imports.

In the late 1980s, government expenditure began to exceed its revenue by such large margins that meeting the expenditure through borrowings became unsustainable. Prices of many essential goods rose sharply. Imports grew at a very high rate without matching growth of exports. As pointed out earlier, foreign exchange reserves declined to a level that was not adequate to finance imports for more than two weeks. There was also not sufficient foreign exchange to pay the interest that needed to be paid to international lenders. Also no country or international funder was willing to lend to India.

India approached the **International Bank for Reconstruction and Development (IBRD)**, popularly known as **World Bank** and the **International Monetary Fund (IMF)**, and received $7 billion as loan to manage the crisis. For availing the loan, these international agencies expected India to liberalise and open up the economy by removing restrictions on the private sector, reduce the role of the government in many areas and remove trade restrictions between India and other countries.

India agreed to the conditionalities of World Bank and IMF and announced the **New Economic Policy (NEP)**. The NEP consisted of wide ranging economic reforms. The thrust of the policies was towards creating a more competitive environment in the economy and removing the barriers to entry and growth of firms. This set of policies can broadly be classified into two groups: the stabilisation measures and the structural reform measures. Stabilisation measures are short-term measures, intended to correct some of the weaknesses that have developed in the **balance of payments** and to bring **inflation** under control. In simple words, this means that there was a need to maintain sufficient foreign exchange reserves and keep the rising prices under control. On the other hand, structural reform policies are long-term measures, aimed at improving the efficiency of the economy and increasing its international competitiveness by removing the rigidities in various segments of the Indian economy. The government initiated a variety of policies which fall under three heads

Reprint 2026-27

viz., liberalisation, privatisation and globalisation.

3.3 Liberalisation

As pointed out in the beginning, rules and laws which were aimed at regulating the economic activities became major hindrances in growth and development. Liberalisation was introduced to put an end to these restrictions and open various sectors of the economy. Though a few liberalisation measures were introduced in 1980s in areas of industrial licensing, **export-import policy**, technology upgradation, **fiscal policy** and foreign investment, reform policies initiated in 1991 were more comprehensive. Let us study some important areas, such as the industrial sector, financial sector, tax reforms, **foreign exchange markets** and trade and investment sectors which received greater attention in and after 1991.

Deregulation of Industrial Sector: In India, regulatory mechanisms were enforced in various ways (i) industrial licensing under which every entrepreneur had to get permission from government officials to start a firm, close a firm or decide the amount of goods that could be produced (ii) private sector was not allowed in many industries (iii) some goods could be produced only in small-scale industries, and (iv) controls on price fixation and distribution of selected industrial products.

The reform policies introduced in and after 1991 removed many of these restrictions. Industrial licensing was abolished for almost all but product categories — alcohol, cigarettes, hazardous chemicals, industrial explosives, electronics, aerospace and drugs and pharmaceuticals. The only industries which are now reserved for the public sector are a part of atomic energy generation and some core activities in railway transport. Many goods produced by small-scale industries have now been **dereserved**. In most industries, the market has been allowed to determine the prices.

Financial Sector Reforms: Financial sector includes financial institutions, such as commercial banks, investment banks, **stock exchange** operations and foreign exchange market. The financial sector in India is regulated by the Reserve Bank of India (RBI). You may be aware that all banks and other financial institutions in India are regulated through various norms and regulations of the RBI. The RBI decides the amount of money that the banks can keep with themselves, fixes interest rates, nature of lending to various sectors, etc. One of the major aims of financial sector reforms is to reduce the role of RBI from regulator to facilitator of financial sector. This means that the financial sector may be allowed to take decisions on many matters without consulting the RBI.

Reprint 2026-27

The reform policies led to the establishment of private sector banks, Indian as well as foreign. Foreign investment limit in banks was raised to around 74 per cent. Those banks which fulfil certain conditions have been given freedom to set up new branches without the approval of the RBI and rationalise their existing branch networks. Though banks have been given permission to generate resources from India and abroad, certain managerial aspects have been retained with the RBI to safeguard the interests of the account-holders and the nation. **Foreign Institutional Investors (FII),** such as merchant bankers, mutual funds and pension funds, are now allowed to invest in Indian financial markets.

Tax Reforms: Tax reforms are concerned with the reforms in the government's taxation and public expenditure policies, which are collectively known as its **fiscal policy**. There are two types of taxes: direct and indirect. **Direct taxes** consist of taxes on incomes of individuals, as well as, profits of business enterprises. Since 1991, there has been a continuous reduction in the taxes on individual incomes as it was felt that high rates of income tax were an important reason for tax evasion. It is now widely accepted that moderate rates of income tax encourage savings and voluntary disclosure of income. The rate of **corporation tax**, which was very high earlier, has been gradually reduced. Efforts have also been made to reform the **indirect taxes**, taxes levied on goods and services to facilitate the establishment of a common national market.

In the year 2016, Indian constitution was amended to empower state governments and union Government to come out with laws to impose Goods and Services Tax. This has led to introduction of GST in India. This is expected to generate additional revenue for the government, reduce tax evasion and create 'one nation, one tax and one market'. Another component of reform in this area is simplification. In order to encourage better compliance on the part of taxpayers, many procedures have been simplified and the rates also substantially lowered.

Foreign Exchange Reforms: The first important reform in the external sector was made in the foreign exchange market. In 1991, as an immediate measure to resolve the balance of payments crisis, the rupee was **devalued** against foreign currencies. This led to an increase in the inflow of foreign exchange. It also set the tone to free the determination of rupee value in the foreign exchange market from government control. Now, more often than not, markets determine exchange rates based on the demand and supply of foreign exchange.

Trade and Investment Policy Reforms: Liberalisation of trade and investment regime was initiated to increase international competitiveness of industrial production and also foreign investments and technology into the economy. The aim was also to promote the efficiency of local industries and adoption of modern technologies.

Reprint 2026-27

In order to protect domestic industries, India was following a regime of **quantitative restrictions** on imports. This was encouraged through tight control over imports and by keeping the **tariffs** very high. These policies reduced efficiency and competitiveness which led to slow growth of the manufacturing sector. The trade policy reforms aimed at (i) dismantling of quantitative restrictions on imports and exports (ii) reduction of tariff rates and (iii) removal of licensing procedures for imports. **Import licensing** was abolished except in case of hazardous and environmentally sensitive industries. Quantitative restrictions on imports of manufactured consumer goods and agricultural products were also fully removed from April 2001. **Export duties** have been removed to increase the competitive position of Indian goods in the international markets.



Work These Out

- Give an example each of nationalised bank, private bank, private foreign bank, FII and a mutual fund.
- Visit a bank in your locality with your parents. Observe and find out the functions it performs. Discuss the same with your classmates and prepare a chart on it.
- Find out from your parents if they pay taxes. If yes, why do they do so and how?
- Do you know that for a very long time countries used to keep silver and gold as reserves to make payments abroad? Find out in what form do we keep our foreign exchange reserves and find out from newspapers, magazines and the Economic Survey how much foreign exchange reserves India had during the last year. Also find the foreign currency of the following countries and its latest rupee exchange rate.

Country	*Currency*	*Value of 1(one) unit of foreign currency in Indian rupee*
U.S.A. U.K. Japan China Korea Singapore Germany		

© NCERT not to be republished

3.4 Privatisation

It implies shedding of the ownership or management of a government owned enterprise. Government companies are converted into private companies in two ways (i) by withdrawal of the government from ownership and management of public sector companies and or (ii) by outright sale of public sector companies.

Privatisation of the public sector enterprises by selling off part of the equity of PSEs to the public is known as **disinvestment**. The purpose of the sale, according to the government, was mainly to improve financial discipline and facilitate modernisation. It was also envisaged that private capital and managerial capabilities could be effectively utilised to improve the performance of the PSUs.

Box 3.1: *Navratnas* and Public Enterprise Policies

You must have read in your childhood about the famous *Navratnas* or Nine Jewels in the Imperial Court of King Vikramaditya who were eminent persons of excellence in the fields of art, literature and knowledge. In order to improve efficiency, infuse professionalism and enable them to compete more effectively in the liberalised global environment, the government identifies PSEs and declare them as *maharatnas*, *navratnas* and *miniratnas*. They were given greater financial, managerial and operational autonomy, in taking various decisions to run the company efficiently and thus increase their profits.

The Central Public Sector Enterprises are designated with different status. A few examples of public enterprises with their status are as follows: (i) *Maharatnas* – (a) Indian Oil Corporation Limited, and (b) Steel Authority of India Limited, (ii) *Navratnas* – (a) Hindustan Aeronautics Limited, (b) Mahanagar Telephone Nigam Limited; and (c) Indian Railway Catering and Tourism Corporation Limited, and (iii) *Miniratnas* – (a) Bharat Sanchar Nigam Limited; (b) Airport Authority of India.

Many of these profitable PSEs were originally formed during the 1950s and 1960s when self-reliance was an important element of public policy. They were set up with the intention of providing infrastructure and direct employment to the public so that quality end-product reaches the masses at a nominal cost and the companies themselves were made accountable to all stakeholders.

The granting of status resulted in better performance of these companies. Scholars allege that instead of facilitating public enterprises in their expansion and enabling them to become global players, the government partly privatised them through disinvesment. Of late, the government has decided to retain them in the public sector and enable them to expand themselves in the global markets and raise resources by themselves from financial markets.

Reprint 2026-27



Work These Out

- Some scholars refer to disinvestment as the wave of privatisation spreading all over the world to improve the performance of public sector enterprises whereas others call it as outright sale of public property to the vested interests. What do you think?
- Prepare a poster which contains 10-15 news clippings which you consider as important and relating to *navaratnas, maharatnas, and miniratnas* from newspapers. Also collect the logos and advertisements of these PSEs. Put these on the notice board and discuss them in the classroom.
- Do you think only loss making companies should be privatised? Why?
- Losses incurred by public sector enterprises are to be met out of the public budget. Do you agree with this statement? Discuss.

The government envisaged that privatisation could provide strong impetus to the inflow of FDI.

The government has also made attempts to improve the efficiency of PSUs by giving them autonomy in taking managerial decisions. For instance, some PSUs have been granted special status as *maharatnas, navratnas* and *miniratnas* (see Box 3.1).

3.5 Globalisation

Although globalisation is generally understood to mean integration of the economy of the country with the world economy, it is a complex phenomenon. It is an outcome of the set of various policies that are aimed at transforming the world towards greater interdependence and integration. It involves creation of networks and activities transcending economic, social and geographical boundaries. Globalisation attempts to establish links in such a way that the happenings in India can be influenced by events happening miles away. It is turning the world into one whole or creating a borderless world.

Outsourcing: This is one of the important outcomes of the globalisation process. In outsourcing, a company hires regular service from external sources, mostly from other countries, which was previously provided internally or from within the country (like legal advice, computer service, advertisement, security — each provided by respective departments of the company). As a form of economic activity, outsourcing has intensified, in recent times, because of the growth of fast modes of communication, particularly the growth of Information Technology (IT). Many of the services such as voice-based business processes (popularly known as BPO or call centres), record keeping,

Reprint 2026-27

Box 3.2: Global Footprint!

Owing to globalisation, you might find many Indian companies have expanded their wings to many other countries. For example, ONGC Videsh, a subsidiary of the Indian public sector enterprise, Oil and Natural Gas Corporation engaged in oil and gas exploration and production has projects in 16 countries. Tata Steel, a private company established in 1907, is one of the top ten global steel companies in the world which have operations in 26 countries and sell its products in 50 countries. It employs nearly 50,000 persons in other countries. HCL Technologies, one of the top five IT companies in India has offices in 31 countries and employs about 15,000 persons abroad. Dr Reddy's Laboratories, initially was a small company supplying pharmaceutical goods to big Indian companies, today has manufacturing plants and research centres across the world.

Source: *www.rediff.com accessed on 14.10.2014.*

accountancy, banking services, music recording, film editing, book transcription, clinical advice or even teaching are being outsourced by companies in developed countries to India. With the help of modern telecommunication links including the Internet, the text, voice and visual data in respect of these services is digitised and transmitted in real time over continents and national boundaries. Most multinational corporations, and even small companies, are outsourcing their services to India where they can be availed at a cheaper cost with reasonable degree of skill and accuracy. The low wage rates and availability of skilled manpower in India have made it a destination for global outsourcing in the post-reform period.

World Trade Organisation (WTO): The WTO was founded in 1995 as the successor organisation to the General Agreement on Trade and Tariff (GATT). GATT was established in 1948 with 23 countries as the global trade organisation to administer all **multilateral trade agreements** by providing equal opportunities to all countries in the international market for trading purposes. WTO is expected to establish a rule-based trading regime in which nations cannot place arbitrary restrictions on trade. In addition, its purpose is also to enlarge



Fig. 3.1 *Outsourcing: a new employment opportunity in big cities*

Reprint 2026-27

Work These Out

- Many scholars argue that globalisation is a threat as it reduces the role of the government in many sectors. Some counter argue that it is an opportunity as it opens up markets to compete in and capture. Debate in the classroom.
- Prepare a chart consisting of a list of five companies that have BPO services in India, along with their turnover.
- Did you attend online classes or watched videos of your teachers or any other teacher taking classes during the last year through television, mobile phone or computers due to Covid 19 Pandemic? Share your experiences related to information technology.
- Is employment in call centres sustainable? What kinds of skills should people working in call centres acquire to get a regular income?
- If the multinational companies outsource many services to countries like India because of cheap manpower, what will happen to people living in the countries where the companies are located? Discuss.

production and trade of services, to ensure optimum utilisation of world resources and to protect the environment. The WTO agreements cover trade in goods as well as services to facilitate international trade (**bilateral** and multilateral) through removal of tariff as well as **non-tariff barriers** and providing greater market access to all member countries.

As an important member of WTO, India has been in the forefront of framing fair global rules, regulations and safeguards and advocating the interests of the developing world. India has kept its commitments towards liberalisation of trade, made in the WTO, by removing quantitative restrictions on imports and reducing tariff rates.

TABLE 3.1
Growth of GDP and Major Sectors (in %)

Sector	*1980-91*	*1992-2001*	*2002-07*	*2007-12*	*2012-13*	*2013-14*	*2021-22*
Agriculture	3.6	3.3	2.3	3.2	1.5	4.2	4.8*
Industry	7.1	6.5	9.4	7.4	3.6	5	12.7*
Services	6.7	8.2	7.8	10	8.1	7.8	9.2*
Total	**5.6**	**6.4**	**7.8**	**8.2**	**5.6**	**6.6**	**9.4**

Source: *Economic Survey for various years, Ministry of Finance, Government of India.*

Note: **Data pertaining to Gross Value Added (GVA). The GVA is estimated from GDP by adding subsidies on production and subtracting indirect taxes.*

Reprint 2026-27

Some scholars question the usefulness of India being a member of the WTO as a major volume of international trade occurs among the developed nations. They also say that while developed countries file complaints over agricultural subsidies given in their countries, developing countries feel cheated as they are forced to open their markets for developed countries but are not allowed access to the markets of developed countries. What do you think?



Fig. 3.2 *IT industry is seen as a major contributor to India's exports*

3.6 Indian Economy During Reforms: An Assessment

The reform process has completed three decades since its introduction. Let us now look at the performance of the Indian economy during this period. In economics, the growth of an economy is measured by the Gross Domestic Product. Look at Table 3.1. The post–1991 India witnessed a rapid growth in GDP on a continual basis for two decades. The growth of GDP increased from 5.6 per cent during 1980–91 to 9.4 per cent during 2021–22. During the reform period, the growth of agriculture has declined. While the industrial sector reported fluctuation, the growth of the service sector has gone up. This indicates that GDP growth is mainly driven by growth in the service sector. During 2012-13, there has been a setback in the growth rates of different sectors witnessed post–1991. While agriculture recorded a high growth rate during 2013–14, this sector witnessed small growth in 2021–22. The service sector continued to witness a high level of growth — higher than the overall GDP growth during 2007–22. The industrial sector witnessed a steep decline during 2012–13, in the subsequent years it began to show a continuous positive growth.

The opening of the economy has led to a rapid increase in foreign direct investment and foreign exchange reserves. The foreign investment, which includes **foreign direct investment (FDI)** and **foreign institutional investment (FII)**, has increased from about US $100 million in 1990–91 to US $ 23 billion in 2022–23. There has been an increase in the foreign exchange reserves from about US $ 6 billion in 1990-91 to about US $ 646 billion in 2023–24. India is one of the largest foreign exchange reserve holders in the world.

Since 1991, India is seen as a successful exporter of auto parts, pharmaceutical goods, engineering goods, IT software and textiles. Rising prices have also been kept under control.

Reprint 2026-27



Work These Out

- In the previous chapter, you might have studied about subsidies in various sectors, including agriculture. Some scholars argue that subsidy in agriculture should be removed to make the sector internationally competitive. Do you agree? If so, how can it be done? Discuss in class.

- Read the following passage and discuss in class.

  Groundnut is a major oilseed crop in Andhra Pradesh. Mahadeva, who was a farmer in Anantpur district of Andhra Pradesh, used to spend Rs. 10,000 for growing groundnut on his plot of half an acre. The cost included expenditure on raw material (seeds, fertilisers, etc.), labour, bullock power and machinery used. On an average, Mahadeva used to get two quintals of groundnut, and each quintal was sold for Rs. 7,000. Mahadeva, thus, was spending Rs. 10,000 and getting an income of Rs. 14,000. Anantpur district is a drought-prone area. As a result of economic reforms, the government did not undertake any major irrigation project. Recently, groundnut crop in Anantpur is facing problems due to crop disease. Research and extension work has gone down due to lower government expenditure. Mahadeva and his friends brought this matter repeatedly to the notice of the government officials entrusted with this responsibility, but failed. Subsidy was reduced on material (seeds, fertilisers) which increased Mahadeva's cost of cultivation. Moreover, the local markets were flooded with cheap imported edible oils, which was a result of removal of restriction on imports. Mahadeva was not able to sell his groundnut in the market as he was not getting the price to cover his cost.

  What could be done to farmers like Mahadeva from incurring losses? Discuss in the class.

The reform process has been widely criticised for not being able to address some of the basic problems facing our economy especially in areas of employment, agriculture, industry, infrastructure development and fiscal management. The evidences examined so far shown that it is not completely true. What do you think?

Growth and Employment: Though the GDP growth rate has increased in the reform period, scholars point out that the reform-led growth has not generated sufficient employment opportunities in the country. You will study the link between different aspects of employment and growth in the next unit.

Reprint 2026-27

Reforms in Agriculture: Reforms have not been able to benefit agriculture, where the growth rate has been decelerating.

Since 1991, public investment in agriculture sector especially in infrastructure, which includes irrigation, power, roads, market linkages and research and extension (which played a crucial role in the Green Revolution), has fallen. Further, the partial removal of fertiliser subsidy has led to increase in the cost of production, which has severely affected the small and marginal farmers. This sector has been experiencing a number of policy changes such as reduction in import duties on agricultural products, low minimum support price and lifting of quantitative restrictions on the imports of agricultural products. These have adversely affected Indian farmers as they now have to face increased international competition.

Moreover, because of export-oriented policy strategies in agriculture, there has been a shift from production for the domestic market towards production for the export market focusing on cash crops in lieu of production of food grains. This puts pressure on prices of food grains.

Reforms in Industry: Industrial growth has also recorded a slowdown. This is because of decreasing demand of industrial products due to various reasons such as cheaper imports, inadequate investment in infrastructure etc. In a globalised world, developing countries are compelled to open up their economies to greater flow of goods and capital from developed countries and rendering their industries vulnerable to imported goods. Cheaper imports have, thus, replaced the demand for domestic goods. Domestic manufacturers are facing competition from imports. The infrastructure facilities, including power supply, have remained inadequate due to lack of investment. Globalisation is, thus, often seen as creating conditions for the free movement of goods and services from foreign countries that adversely affect the local industries and employment opportunities in developing countries.

Moreover, a developing country like India still does not have the access to developed countries' markets because of high non-tariff barriers. For example, although all quota restrictions on exports of textiles and clothing have been removed in India, USA has not removed their quota restriction on import of textiles from India and China.

Disinvestment: Every year, the government fixes a target for disinvestment of PSEs. For instance, in 1991-92, it was targeted to mobilise Rs 2500 crore through disinvestment. The government was able to mobilise ₹ 3,040 crore more than the target. In 2022-23, government was able to mobilise about Rs. 46,000 crores. Critics

Reprint 2026-27

point out that the assets of PSEs have been undervalued and sold to the private sector. This means that there has been a substantial loss to the government and the outright sale of public assets! Moreover, the proceeds from disinvestment are used to offset the shortage of government revenues rather than using it for the development of PSEs and building social infrastructure in the country. Do you think selling a part of the properties of government companies is the best way to improve their efficiency?

Reforms and Fiscal Policies: Economic reforms have placed limits on the growth of public expenditure, especially in social sectors. The tax reductions in the reform period, aimed at yielding larger revenue and curb tax evasion, have not resulted in increase in tax revenue for the government. Also, the reform policies, involving tariff reduction, have curtailed the scope for raising revenue through custom duties. In order to attract foreign investment, tax incentives are provided to foreign investors which further reduced the scope for raising tax revenues. This has a negative impact on developmental and welfare expenditures.

3.7 Conclusion

The process of globalisation through liberalisation and privatisation policies has produced positive, as well as, negative results both for India and other countries. Some scholars argue that globalisation should be seen as an opportunity in terms of greater access to global markets, high technology and increased possibility of large industries of developing countries to become important players in the international arena.

Box 3.3: Siricilla Tragedy!

Power sector reforms in many Indian states led to do away with the supply of electricity at subsidised rates and steep rise in power tariff. This has affected workers engaged in small industries. Powerloom textile industry in Andhra Pradesh is an example. Since the wages of the powerloom workers are linked to the production of cloth, power cut means cut in wages of weavers, who were already suffering from hike in tariff. A few years ago, this led to a crisis in the livelihood of the weavers and 50 powerloom workers committed suicide in a small town called 'Siricilla' in Andhra Pradesh.

- Do you think the power tariff should not be raised?
- What would be your suggestions to revive small industries affected by reforms?

Reprint 2026-27

"... the wisdom of expending public and private funds on education is not to be measured by its direct fruits alone. It will be profitable as a mere investment, to give the masses of people much greater opportunities than they can generally avail themselves of. For by this means many, who would have died unknown, are enabled to get the start needed for bringing out their latent abilities".

Alfred Marshall

4.1 INTRODUCTION

Think of one factor that has made a great difference in the evolution of mankind. Perhaps it is human capacity to store and transmit knowledge which they have been doing through conversation, through songs and through elaborate lectures. But humans soon found out that we need a good deal of training and skill to do things efficiently. We know that the labour skill of an educated person is more than that of an uneducated person and hence the former is able to generate more income than the latter and his or her contribution to economic growth is, consequently, more.

Education is sought not only as it confers higher earning capacity on people but also for its other highly valued benefits: it gives one a better social standing and pride; it enables one to make better choices in life; it provides knowledge to understand the changes taking place in society; it also stimulates innovations. Moreover, the availability of educated labour force facilitates adaptation of new



Fig. 4.1 *Adequate education and training to farmers can raise productivity in farms*

Reprint 2026-27

technologies. Economists have stressed the need for expanding educational opportunities in a nation as it accelerates the development process.

4.2 What is Human Capital?

Just as a country can turn physical resources like land into physical capital like factories, similarly, it can also turn human resources like nurses, farmers, teachers, students into human capital like engineers and doctors. Societies need sufficient human capital in the first place—in the form of competent people who have themselves been educated and trained as professors and other professionals. In other words, we need good human capital to produce other human capital (say, nurses, farmers, teachers, doctors, engineers...). This means that we need investment in human capital to produce more human capital out of human resources.

Let us understand a little more of what human capital means by posing the following questions :

(i) What are the sources of human capital?
(ii) Is there any relation between human capital and economic growth of a country?
(iii) Is the formation of human capital linked to people's all-round development or, as it is now called, human development?
(iv) What role can the government play in human capital formation in India?

4.3 Sources of Human Capital

Investment in education is considered as one of the main sources of human capital. There are several other sources as well. Investments in health, on-the-job training, migration and information are the other sources of human capital formation.

Why do your parents spend money on education? Spending on education by individuals is similar to spending on capital goods by companies with the objective of increasing future profits over a period of time. Likewise, individuals invest in education with the objective of increasing their future income.

Work This Out

- Identify and collect the data from three families from different strata (i) very poor (ii) middle class and (iii) affluent. Study the expenditure pattern of the families on education of male and female children.

Like education, health is also considered as an important input for the development of a nation as much as it is important for the development of an individual.

Who can work better—a sick person or a person with sound health? A sick labourer without access to medical facilities is compelled to abstain from work and there is loss of productivity. Hence, expenditure on health is an important source of human capital formation.

Reprint 2026-27



The amount of money spent on preventive medicine (vaccination), **curative medicine** (medical intervention during illness), **social medicine** (spread of health literacy) and provision of clean drinking water and good sanitation are the various forms of health expenditures. Health expenditure directly increases the supply of healthy labour force and is, thus, a source of human capital formation.



Firms spend on giving **on-the-job-training** to their workers. This may take different forms: one, the workers may be trained in the firm itself under the supervision of a skilled worker; two, the workers may be sent for off-campus training. In both these cases firms incur some expenses. Firms will, thus, insist that the workers should work for a specific period of time, after their on-the-job training, during which it can recover the benefits of the enhanced productivity owing to the training. Expenditure regarding on-the-job training is a source of human capital formation as the return of such expenditure in the form of enhanced labour productivity is more than the cost of it.

People **migrate** in search of jobs that fetch them higher salaries than what they may get in their native



places. Unemployment is the reason for the rural-urban migration in India. Technically qualified persons, like engineers and doctors, migrate to other countries because of higher salaries that they may get in such countries. Migration in both these cases involves cost of transport, higher cost of living in the migrated places and psychic costs of living in a strange socio-cultural setup. The enhanced earnings in the new place outweigh the costs of migration; hence, expenditure on migration is also a source of human capital formation.

People spend to acquire **information** relating to the labour market and other markets like education and health. For example, people want to know the level of salaries associated with various types of jobs, whether the educational institutions provide the right type of employable skills and at what cost. This information is necessary to make decisions regarding investments in human capital as well as for efficient utilisation of the acquired human capital stock. Expenditure incurred for acquiring information relating to the labour market and other markets is also a source of human capital formation.

Reprint 2026-27

Box 4.1: Physical and Human Capital

Both the forms of capital formation are outcomes of conscious investment decisions. Decision regarding investment in physical capital is taken on the basis of one's knowledge in this regard. The entrepreneur possesses knowledge to calculate the expected rates of return to a range of investments and then rationally decides which one of the investments should be made. The ownership of physical capital is the outcome of the conscious decision of the owner — the physical capital formation is mainly an economic and technical process. A substantial part of the human capital formation takes place in one's life when she/he is unable to decide whether it would maximise her/his earnings. Children are given different types of school education and health care facilities by their parents and the society. The peers, educators and society influence the decisions regarding human capital investments even at the tertiary level, that is, at the college level. Moreover, the human capital formation at this stage is dependent upon the already formed human capital at the school level. Human capital formation is partly a social process and partly a conscious decision of the possessor of the human capital.

You know that the owner of a physical capital, say a bus, need not be present in the place where it is used; whereas, a bus-driver, who possesses the knowledge and ability to drive the bus, should be present when the bus is used for transportation of people and materials. Physical capital is tangible and can be easily sold in the market like any other commodity. Human capital is intangible; it is endogenously built in the body and mind of its owner. Human capital is not sold in the market; only the services of the human capital are sold and, hence, there arises the necessity of the owner of the human capital to be present in the place of production. The physical capital is separable from its owner, whereas, human capital is inseparable from its owner.

The two forms of capital differ in terms of mobility across space. Physical capital is completely mobile between countries except for some artificial trade restrictions. Human capital is not perfectly mobile between countries as movement is restricted by nationality and culture. Therefore, physical capital formation can be built even through imports, whereas human capital formation is to be done through conscious policy formulations in consonance with the nature of the society and economy and expenditure by the state and the individuals.

Both forms of capital depreciate with time but the nature of depreciation differs between the two. Continuous use of machine leads to depreciation and change of technology makes a machine obsolete. In the case of human capital, depreciation takes place with ageing but can be reduced, to a large extent, through continuous investment in education, health, etc. This investment also facilitates the human capital to cope with change in technology which is not the case with physical capital.

Nature of benefits flowing from human capital are different from that of physical capital. Human capital benefits not only the owner but also the society in general. This is called external benefit. An educated person can effectively take part in a democratic process and contribute to the socio-economic progress of a nation. A healthy person, by maintaining personal hygiene and sanitation, stops the spread of contagious diseases and epidemics. Human capital creates both private and social benefits, whereas physical capital creates only private benefit. That is, benefits from a capital good flow to those who pay the price for the product and services produced by it.

Reprint 2026-27

The concept of physical capital is the base for conceptualising human capital. There are some similarities between the two forms of capital; there are some striking dissimilarities as well. See Box 4.1.

Human Capital and Economic Growth: Who contributes more to national income — a worker in a factory or a software professional? We know that the labour skill of an educated person is more than that of an uneducated person and that the former generates more income than the latter. Economic growth means the increase in real national income of a country; naturally, the contribution of the educated person to economic growth is more than that of an illiterate person. If a healthy person could provide uninterrupted labour supply for a longer period of time, then health is also an important factor for economic growth. Thus, both education and health, along with many other factors like on-the-job training, job market information and migration, increase an

Look at Fig. 4.2 and discuss.

(a) What are the advantages of having proper 'classroom'?

(b) Do you think the children going to this schol are receiving quality education?

(c) Why these schools do not have buildings?

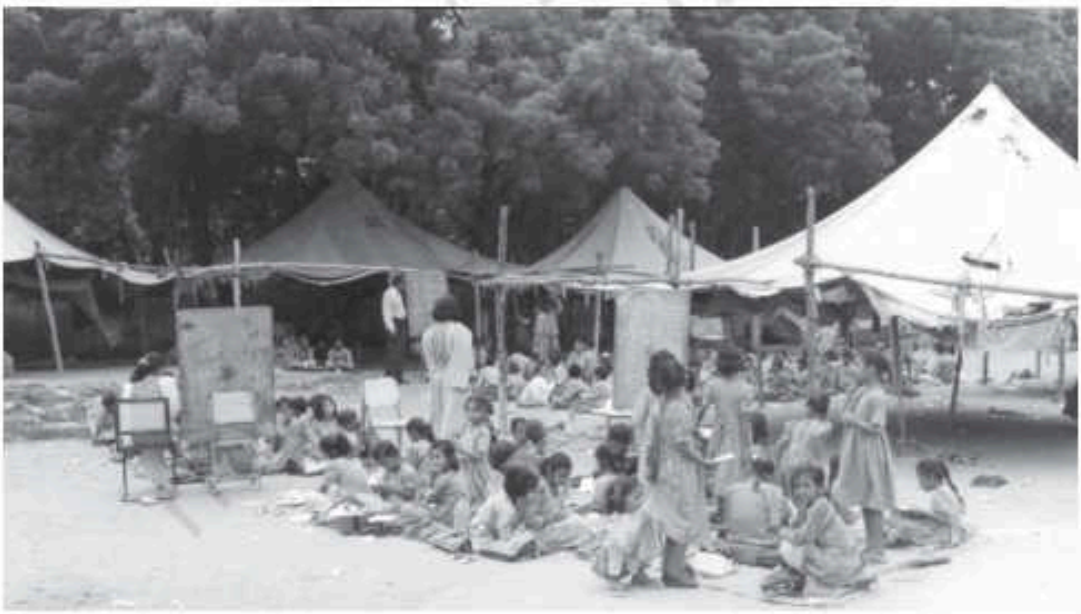


Fig. 4.2 *Creating human capital: a school being run in make shift premises in Delhi*

Reprint 2026-27

individual's income generating capacity.

This enhanced productivity of human beings or human capital contributes substantially not only towards increasing labour productivity but also stimulates innovations and creates ability to absorb new technologies. Education provides knowledge to understand changes in society and scientific advancements, thus, facilitate inventions and innovations. Similarly, the availability of educated labour force facilitates adaptation to new technologies.

Empirical evidence to prove that increase in human capital causes economic growth is rather nebulous. This may be because of measurement problems. For example, education measured in terms of years of schooling, teacher-pupil ratio and enrolment rates may not reflect the quality of education; health services measured in monetary terms, life expectancy and mortality rates may not reflect the true health status of the people in a country. Using the indicators mentioned above, an analysis of improvement in education and health sectors and growth in real per capita income in both developing and developed countries shows that there is convergence in the measures of human capital but no sign of convergence of per capita real income. In other words, the human capital growth in developing countries has been faster but the growth of per capita real income has not been that fast. There are reasons to believe that the



Fig. 4.3 *Scientific and technical manpower: a rich ingredient of human capital*

causality between human capital and economic growth flows in either directions. That is, higher income causes building of high level of human capital and vice versa, that is, high level of human capital causes growth of income.

India recognised the importance of human capital in economic growth long ago. The Seventh Five Year Plan says, "Human resources development (read human capital) has necessarily to be assigned a key role in any development strategy, particularly in a country with a large population. Trained and educated on sound lines, a large population can itself become an asset in accelerating economic growth and in ensuring social change in desired directions."

It is difficult to establish a relation of cause and effect from the growth of human capital (education and health) to economic growth but we can see in

Reprint 2026-27

TABLE 4.1

Select Indicators of Development in Education and Health Sectors

Particulars		*1951*	*1981*	*1991*	*2001*	*2018-22*
Real Per Capita Income (in Rs)		7,651	12,174	15,748	23,095	94,054
Crude Death Rate (Per 1,000 Population)		25.1	12.5	9.8	8.1	6.0
Infant Mortality Rate		146	110	80	63	28
Life Expectancy at Birth (in Years)	Male	37.2	54.1	59.7	63.9	68.6
	Female	36.2	54.7	60.9	66.9	71.4
Literacy Rate (%)		16.67	43.57	52.21	65.20	78

Source: *Economic Survey for various years, Ministry of Finance; National Statistical Office, Ministry of Statistics and Programme Implementation, Government of India.*

Table 4.1 that these sectors have grown simultaneously. Growth in each sector probably has reinforced the growth of every other sector.

The National Education Policy 2020 states that the world is undergoing rapid changes in the knowledge landscape. With various dramatic scientific and technological advances, such as the rise of big data, machine learning, and artificial intelligence, many unskilled jobs worldwide may be taken over by machines, while the need for a skilled workforce, particularly involving mathematics, computer science, and data science, in conjunction with multidisciplinary abilities across the sciences and social sciences, and humanities, will be increasingly in greater demand. With climate change, increasing pollution, and depleting natural resources, there will be a sizeable shift in how we meet the world's energy, water, food, and sanitation needs, again resulting in the need for new skilled labour, particularly in biology, chemistry, physics, agriculture, climate science, and social science. The growing emergence of epidemics and pandemics will also call for collaborative research in infectious disease management and development of vaccines and the resultant social issues heightens the need for multidisciplinary learning. There will be a growing demand for humanities and art, as India moves towards becoming a developed country as well as



Fig. 4.4 *Job on hand: transforming India into a knowledge economy*

Reprint 2026-27

Box 4.2: India as a Knowledge Economy

The Indian software industry has been showing an impressive record over the past two decades. Entrepreneurs, bureaucrats and politicians are now advancing views about how India can transform itself into a knowledge-based economy by using information technology (IT). There have been some instances of villagers using e-mail which are cited as examples of such transformation. Likewise, e-governance is being projected as the way of the future. The value of IT depends greatly on the existing level of economic development. Do you think IT – based services in rural areas will lead to human development? Discuss.

among the three largest economies in the world. This policy vision suggests how human capital formation in India will move its economy to a higher growth trajectory based on knowledge landscape.

4.4 Human Capital and Human Development

The two terms sound similar but there is a clear distinction between them. Human capital considers education and health as a means to increase labour productivity. Human development is based on the idea that education and health are integral to human well-being because only when people have the ability to read and write and the ability to lead a long and healthy life, they will be able to make other choices which they value. Human capital treats human beings as a means to an end; the end being the increase in productivity. In this view, any investment in education and health is unproductive if it does not enhance output of goods and services. In the human development perspective, human beings are ends in themselves. Human welfare should be increased through investments in education and health even if such investments do not result in higher labour productivity. Therefore, basic education and basic health are important in themselves, irrespective of their contribution to labour productivity. In such a view, every individual has a right to get basic education and basic health care, that is, every individual has a right to be literate and lead a healthy life.



Work This Out

- If a construction worker, maid-servant, *dhobi* or a peon in school has absented herself/himself for long due to ill health, find out how it has affected her/his
  - (i) job security
  - (ii) wage/salary

What could be the possible reasons?

Reprint 2026-27

4.5 State of Human Capital Formation in India

In this section we are going to analyse human capital formation in India. We have already learnt that human capital formation is the outcome of investments in education, health, on-the-job training, migration and information. Of these education and health are very important sources of human capital formation. We know that India is a federal country with a union government, state governments and local governments (Municipal Corporations, Municipalities and Village Panchayats). The Constitution of India mentions the functions to be carried out by each level of government. Accordingly, expenditures on both education and health are to be carried out simultaneously by all the three tiers of the government. Analysis of health sector is taken up in Chapter 8; hence, we will analyse only the education sector here.

Do you know who takes care of education and health in India? Before we take up the analysis of the education sector in India, we will look into the need for government intervention in education and health sectors. We do understand that education and health care services create both private and social benefits and this is the reason for the existence of both private and public institutions in the education and health service markets. Expenditures on education and health make substantial long-term impact and they cannot be easily reversed; hence, government intervention is essential. For instance, once a child is admitted to a school or health care centre where the required services are not provided, before the decision is taken to shift the child to another institution, substantial amount of damage would have been done. Moreover, individual consumers of these services do not have complete information about the quality of services and their costs. In this situation, the providers of education and health services acquire monopoly power and are involved in exploitation. The role of government in this situation is to ensure that the private providers of these services adhere to the standards stipulated by the government and charge the correct price.

In India, the ministries of education at the union and state level, departments of education and various organisations like National Council of Educational Research and Training (NCERT), University Grants Commission (UGC) and All India Council of Technical Education (AICTE) facilitate institutions which come under the education sector. Similarly, the ministries of health at the union and state level, departments of health and various organisations like National Medical Commission and Indian Council for Medical Research (ICMR) facilitate institutions which come under the health sector.

In a developing country like India, with a large section of the population living below the poverty line, many people cannot afford to access basic education and health care facilities. Moreover, a substantial section of India's population cannot afford

Reprint 2026-27

to reach super specialty health care and higher education. Furthermore, when basic education and health care is considered as a right of the citizens, then it is essential that the government should provide education and health services free of cost for the deserving citizens and those from the socially oppressed classes. Both, the union and state governments, have been stepping up expenditures in the education sector over the years in order to fulfil the objective of attaining cent per cent literacy and considerably increase the average educational attainment of Indians.



Work This Out

- Identify the objectives and functions of NCERT, UGC, AICTE and ICMR.

4.6 Education Sector in India

Growth in Government Expenditure on Education: Do you know how much the government spends on education? This expenditure by the government is expressed in two ways (i) as a percentage of 'total government expenditure' (ii) as a percentage of Gross Domestic Product (GDP).

The percentage of 'education expenditure of total government expenditure' indicates the importance of education in the scheme of things before the government. The percentage of 'education expenditure of GDP' expresses how much of people's income is being committed to the development of education in the country. During 1952-2020, education expenditure as percentage of total government expenditure increased from 7.92 to 16.54 and as percentage of GDP increased from 0.64 to 4.47. Throughout this period the increase in education expenditure has not been uniform and there has been irregular rise and fall. To this if we include the private expenditure incurred by individuals and by philanthropic institutions, the total education expenditure should be much higher.

Elementary education takes a major share of total education expenditure and the share of the higher/tertiary education (institutions of higher learning like colleges, polytechnics and universities) is the least. Though, on an average, the government spends less on tertiary education, 'expenditure per student' in tertiary education is higher than that of elementary. This does not mean that financial resources should be transferred from tertiary education to elementary education. As we expand school education, we need more teachers who are trained in the higher educational institutions; therefore, expenditure on all levels of education should be increased.

In 2020-21, the per capita public expenditure on elementary education differs considerably across states from as high as Rs 96,968 in Sikkim to as low as Rs 10,710 in Bihar. This leads to differences in educational opportunities and attainments across states.

Reprint 2026-27



Fig. 4.5 *Investment in educational infrastructure is inevitable*

One can understand the inadequacy of the expenditure on education if we compare it with the desired level of education expenditure as recommended by the various commissions. The Education Commission (1964–66) had recommended that at least 6 per cent of GDP be spent on education so as to make a noticeable rate of growth in educational achievements. The Tapas Majumdar Committee, appointed by the Government of India in 1999, estimated an expenditure of around Rs 1.37 lakh crore over 10 years (1998-99 to 2006-07) to bring all Indian children in the age group of 6-14 years under the purview of school education. Compared to this desired level of education expenditure of around 6 per cent of GDP, the current level of a little over 4 per cent has been quite inadequate. In principle, a goal of 6 per cent needs to be reached—this has been accepted as a must for the coming years. In 2009, the Government of India enacted the Right of Children to Free and Compulsory Education Act to make free education a fundamental right of all children in the age group of 6-14 years.

Government of India has also started levying a 2 per cent 'education cess' on all Union taxes. The revenues from education cess has been earmarked for spending on elementary education. In addition to this, the government sanctions a large outlay for the promotion of higher education and

Work These Out

- Prepare case studies of dropouts at different levels of schooling, say
  - (i) Primary dropouts
  - (ii) Class VIII dropouts
  - (iii) Class X dropouts

  Find out the causes and discuss in the class.
- 'School dropouts are giving way to child labour'. Discuss how this is a loss to human capital.

Reprint 2026-27

TABLE 4.2

Educational Attainment in India

Sl.No.	*Particulars*	*1990*	*2000*	*2011*	*2017-18*
1.	Adult Literacy Rate (per cent of people aged 15+)				
	1.1 Male	61.9	68.4	79	82
	1.2 Female	37.9	45.4	59	66
2.	Primary completion rate (per cent of relevant age group)				
	2.1 Male	78	85	92	93
	2.2 Female	61	69	94	96
3.	Youth literacy rate (per cent of people aged 15+ to 24)				
	3.1 Male	76.6	79.7	90	93
	3.2 Female	54.2	64.8	82	90

new loan schemes for students to pursue higher education.

Educational Achievements in India: Generally, educational achievements in a country are indicated in terms of adult literacy level, primary education completion rate and youth literacy rate. These statistics for the last two decades are given above in Table 4.2.

4.7 Future Prospects

Education for All — Still a Distant Dream: Though literacy rates for both — adults as well as youth — have increased, still the absolute number of illiterates in India is as much as India's population was at the time of independence. In 1950, when the Constitution of India was passed by the Constituent Assembly, it was noted in the Directive Principles of the



Fig. 4.6 *School dropouts give way to child labour: a loss to human capital*

Reprint 2026-27

Constitution that the government should provide free and compulsory education for all children up to the age of 14 years within 10 years from the commencement of the Constitution. Had we achieved this, we would have cent per cent literacy by now.

Gender Equity — Better than Before: The differences in literacy rates between males and females are narrowing signifying a positive development in gender equity; still the need to promote education for women in India is imminent for various reasons such as improving economic independence and social status of women and also because women education makes a favourable impact on fertility rate and health care of women and children. Therefore, we cannot be complacent about the upward movement in the literacy rates and we have miles to go in achieving cent per cent adult literacy.

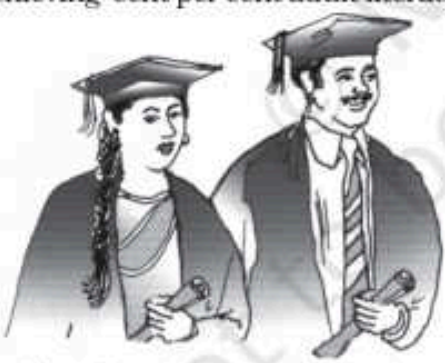


Fig. 4.7 *Higher Education: few takers*

Higher Education — a Few Takers: The Indian education pyramid is steep, indicating lesser and lesser number of people reaching the higher education level. Moreover, the level of unemployment among educated youth is the highest. As per NSSO data, in the year 2011-12, the rate of unemployment among youth males who studied graduation and above in rural areas was 19 per cent. Their urban counterparts had relatively less level of unemployment at 16 per cent. The most severely affected ones were young rural female graduates as nearly 30 per cent of them are unemployed. In contrast to this, only about 3-6 per cent of primary level educated youth in rural and urban areas were unemployed. The situation has improved moderately as indicated by the Periodic Labour Force Survey 2023-24. Nevertheless, the government should increase allocation for higher education and also improve the standard of higher education institutions, so that students are imparted employable skills in such institutions. When compared to less educated, a large proportion of educated persons are unemployed. Why?

4.8 Conclusion

The economic and social benefits of human capital formation and human development are well known. The union and state governments in India have been earmarking substantial financial outlays for development of education and health sectors. The spread of education and health services across different sectors of society should be ensured so as to simultaneously attain economic growth and equity. India has a rich stock of scientific and technical manpower in the world. The need of the hour is to better it qualitatively and provide such conditions so that they are utilised in India.

Reprint 2026-27

activities such as food processing. Enabling them better and more affordable access to healthcare, sanitation facilities at workplaces and homes and education for all would also need to be given top priority for rapid rural development.

It was observed in an earlier chapter that although the share of agriculture sector's contribution to GDP was on a decline, the population dependent on this sector did not show any significant decline. Further, after the initiation of reforms, the growth rate of agriculture sector decelerated to about 3 per cent per annum during the 1991-2012, which was lower than the earlier years. In recent years, this sector has become volatile. During 2023-24, the GVA growth rate of agriculture and its allied sectors was about two per cent. Scholars identify decline in public investment since 1991 as the major reason for this. They also argue that inadequate infrastructure, lack of alternate employment opportunities in the industry or service sector, increasing casualisation of employment etc., further impede rural development. The impact of this phenomenon can be seen from the growing distress witnessed among farmers across different parts of India. Against this background, we will critically look at some of the crucial aspects of rural India like credit and marketing systems, agricultural diversification and the role of organic farming in promoting sustainable development.



Work These Out

- On a monthly basis, go through the newspapers of your region and identify the problems raised by them in relation to rural areas and the solutions offered. You could also visit a nearby village and identify the problems faced by people there. Discuss this in the classroom.
- Prepare a list of recent schemes and their objectives from the government website https://www.rural.nic.in. Collect the details of how anyone of these schemes implemented in your region/rural neighbourhood areas. Discuss your observations in class.

5.3 Credit and Marketing in Rural Areas

Credit: Growth of rural economy depends primarily on infusion of capital, from time to time, to realise higher productivity in agriculture and non-agriculture sectors. As the time of gestation period between crop sowing and realisation of income after production is quite long, farmers borrow from various sources to meet their initial investment on seeds, fertilisers, implements and other family expenses of marriage, death, religious ceremonies etc.

At the time of independence, moneylenders and traders exploited small and marginal farmers and landless labourers by lending to them on high interest rates and by manipulating the accounts to keep them in a debt-trap. A major change occurred after 1969 when India adopted social banking and multi-agency approach to adequately meet the needs of rural credit. Later, the National Bank for Agriculture and

Reprint 2026-27

Box 5.1: The Poor Women's Bank

'Kudumbashree' is a women-oriented community-based poverty reduction programme being implemented in Kerala. In 1995, a thrift and credit society was started as a small savings bank for poor women with the objective to encourage savings. The thrift and credit society mobilised Rs 1 crore as thrift savings. These societies have been acclaimed as the largest informal banks in Asia in terms of participation and savings mobilised.

Source: *www.kudumbashree.org. Visit this website and explore various other initiatives undertaken by this organisation. Identify some factors which contributed to their successes and discuss in the class.*

National Bank for Agriculture and Rural Development (NABARD) was set up in 1982 as an apex body to coordinate the activities of all institutions involved in the rural financing system. The Green Revolution was a harbinger of major changes in the credit system as it led to the diversification of the portfolio of rural credit towards production-oriented lending.

The institutional structure of rural banking today consists of a set of multi-agency institutions, namely, commercial banks, regional rural banks (RRBs), cooperatives and land development banks. They are expected to dispense adequate credit at cheaper rates. Recently, Self-Help Groups (henceforth SHGs) have emerged to fill the gap in the formal credit system because the formal credit delivery mechanism has not only proven inadequate but has also not been fully integrated into the overall rural social and community development. Since some kind of collateral is required, vast proportion of poor rural households were automatically out of the credit network. The SHGs promote thrift in small proportions by a minimum contribution from each member. From the pooled money, credit is given to the needy members to be repayable in small instalments at reasonable interest rates. By May 2019, nearly 6 crore



Work These Out

- In your locality/neighbourhood, you might notice self-help groups providing credit. Attend few meetings of such self-help groups. Write a report on the profile of a self-help group. The profile may include — when it was started, the number of members, amount of savings and type of credit they provide and how borrowers use the loan.
- You might also find that those who take a loan for starting self-employment activities but use it for other purposes. Interact with few such borrowers. Identify the reasons for not starting self employment activities and discuss in the classroom.

Reprint 2026-27

women in India have become member in 54 lakh women SHGs. In recent years, about ₹ 10-15,000 per SHG and another ₹2.5 lakhs per SHG as a Community Investment Support Fund (CISF) are provided as part of renovating fund to take up self employement for income generation. Such credit provisions are generally referred to as **micro-credit programmes**. SHGs have helped in the empowerment of women. It is alleged that the borrowings are mainly confined to consumption purposes. Why are borrowers not spending for productive purposes?

Rural Banking — a Critical Appraisal: Rapid expansion of the banking system had a positive effect on rural farm and non-farm output, income and employment, especially after the green revolution — it helped farmers to avail services and credit facilities and a variety of loans for meeting their prodution needs. Famines became events of the past; we have now achieved food security which is reflected in the abundant buffer stocks of grains. However, all is not well with our banking system.

With the possible exception of the commercial banks, other formal institutions have failed to develop a culture of deposit mobilisation — lending to worthwhile borrowers and effective loan recovery. Agriculture loan default rates have been chronically high. Why farmers failed to pay back loans? It is alleged that farmers are deliberately refusing to pay back loans. What could be the reasons?

Thus, the expansion and promotion of the rural banking sector has taken a backseat after reforms. To improve the situation, In recent years, all the adults are encouraged to open bank accounts as a part of a scheme known as Jan-Dhan Yojana. Those bank account holders can get Rs. 1-2 lakh accidental insurance coverage and overdraft facilities for Rs. 10,000 and get their wages if they get any government-related jobs and works under MNREGA (now known as Viksit Bharat–Guarantee for Rozgar and Ajeevika Mission [Gramin] [VB-G RAM G]); old age pension and



Work These Out

- In the last few years, you might have taken note — in your neighbourhood if you are living in rural areas or read in the newspapers or seen on TV — of farmers commiting suicides. Many such farmers had borrowed money for farming and other purposes. It was found that when they were unable to pay back due to crop failure, insufficient income and employment opportunities, they took such steps. Collect information relating to such cases and discuss in the classroom.
- Visit banks that cater to rural areas. They may be primary agricultural cooperative banks, land development banks, regional rural banks or district cooperative banks. Collect details such as how many rural households borrowed from them, amount generally borrowed, kinds of collateral used, interest rates and dues.
- If farmers who borrowed from cooperative banks could not pay back due to crop failure and other reasons, their loans should be waived otherwise they may take drastic decisions like committing suicides. Do you agree?

Reprint 2026-27

other social security payments of the government are transferred to bank accounts. There is no need to keep minimum bank balance. This has led to more than 50 crore people opening bank accounts; indirectly it has promoted thrift habit and efficient allocation of financial resources particularly in rural areas. Banks also could mobile funds of more than Rs. 2,00,000 crores through these accounts.

5.4 Agricultural Market System

Have you ever asked yourself how food grains, vegetables and fruits that we consume daily come from different parts of the country? The mechanism through which these goods reach different places depends on the market channels. Agricultural marketing is a process that involves the assembling, storage, processing, transportation, packaging, grading and distribution of different agricultural commodities across the country.

Prior to independence, farmers, while selling their produce to traders, suffered from faulty weighing and manipulation of accounts. Farmers who did not have the required information on prices prevailing in markets were often forced to sell at low prices. They also did not have proper storage facilities to keep back their produce for selling later at a better price. Do you know that even today, more than 10 per cent of goods produced in farms are wasted due to lack of storage? Therefore, government intervention became necessary to regulate the activities of the private traders.

Let us discuss four such measures that were initiated to improve the marketing aspect. The first step was regulation of markets to create orderly and transparent marketing conditions. By and large, this policy benefited farmers as well as consumers. However, there is still a need to develop about 27,000 rural periodic markets as regulated market places to realise the full potential of rural markets. Second component is provision of physical infrastructure facilities like roads, railways, warehouses, godowns, cold storages and processing units. The current infrastructure facilities are quite inadequate to meet the growing demand and need to be improved. Cooperative



Fig. 5.1 *Regulated market yards benefit farmers as well as consumers*

Reprint 2026-27



Work These Out

- Visit a nearby vegetable and fruit market. Observe and identify different characteristics of the market. Identify the place of origin of at least ten different fruits and vegetables and distance travelled to reach the market. Further, look at the modes of transport and its implication on prices.
- Most small towns have regulated market yards. Farmers can go to these markets and sell their produce. They can also store their goods in the yard. In 2016, Union Government launched a portal named e-National Agriculture Market (e-NAM), and a Pan-India online portal by connecting *mandis* and RMYs. Visit one regulated market yard or Agriculture Produce Marketing Committee collect the details of its functioning, kind of goods coming to the yard and how prices are fixed. Prepare a detailed report and discuss in the class.

marketing, in realising fair prices for farmers' products, is the third aspect of government initiative. The success of milk cooperatives in transforming the social and economic landscape of Gujarat and some other parts of the country is testimony to the role of cooperatives. However cooperatives have received a setback during the recent past due to inadequate coverage of farmer members, lack of appropriate link between marketing and processing cooperatives and inefficient financial management. The fourth element is the policy instruments like (i) assurance of minimum support prices (MSP) for agricultural products (ii) maintenance of buffer stocks of wheat and rice by Food Corporation of India and (iii) distribution of food grains and sugar through PDS. These instruments are aimed at protecting the income of the farmers and providing foodgrains at a subsidised rate to the poor. However, despite government intervention, private trade (by moneylenders, rural political elites, big merchants and rich farmers) predominates agricultural markets. The need for government intervention is imminent particularly when a large share of agricultural products, is handled by the private sector.

Agricultural marketing has come a long way with the intervention of the government in various forms. Some scholars argue that commercialisation of agriculture offers tremendous scope for farmers to earn higher incomes provided the government intervention is restricted. What do you think about this view?

Emerging Alternate Marketing Channels: It has been realised that if farmers directly sell their produce to consumers, it increases their incomes. Some examples of these channels are *Apni Mandi* (Punjab, Haryana and Rajasthan); *Hadaspar Mandi* (Pune); *Rythu Bazars* (vegetable and fruit markets in Andhra Pradesh and Telangana) and *Uzhavar Sandies* (farmers markets in Tamil Nadu). Further, several national and multinational fast food chains are increasingly entering into contracts/ alliances with farmers to encourage

Reprint 2026-27

them to cultivate farm products (vegetables, fruits, etc.) of the desired quality by providing them with not only seeds and other inputs but also assured procurement of the produce at pre-decided prices. It is argued that such arrangements will help in reducing the price risks of farmers and would also expand the markets for farm products. Do you think such arrangements raise incomes of small farmers.



Work This Out

➢ Visit one such alternative marketing system which farmers in your locality, or in the neighbourhood rural areas, use. How are they different from regulated market yards? Should they be encouraged and supported by the government? Why and how? Discuss.

In 2020, Indian Parliament passed three laws to reform agriculture marketing system. While some section of farmers support these reforms, the rest of the farmers oppose and these Acts were later repealed. Collect the details of these laws, debate and discuss in the class.

5.5 Diversification into Productive Activities

Diversification includes two aspects – one relates to change in cropping pattern and the other relates to a shift of workforce from agriculture to other allied activities (livestock, poultry, fisheries etc.) and non-agriculture sector. The need for diversification arises from the fact that there is greater risk in depending exclusively on farming for livelihood. Diversification towards new areas is necessary not only to reduce the risk from agriculture sector but also to provide productive sustainable livelihood options to rural people. Much of the agricultural employment activities are concentrated in the Kharif season. But during the Rabi season, in areas where there are inadequate irrigation facilities, it becomes difficult to find gainful employment. Therefore expansion into other sectors is essential to provide supplementary gainful employment and in realising higher levels of income for rural people to overcome poverty and other tribulations. Hence, there is a need to focus on allied activities, non-farm employment and other emerging alternatives of livelihood, though there are many other options available for providing sustainable livelihoods in rural areas.



Fig. 5.2 *Jaggery making is an allied activity of the farming sector*

Reprint 2026-27

Box 5.2: Tamil Nadu Women in Agriculture (TANWA)

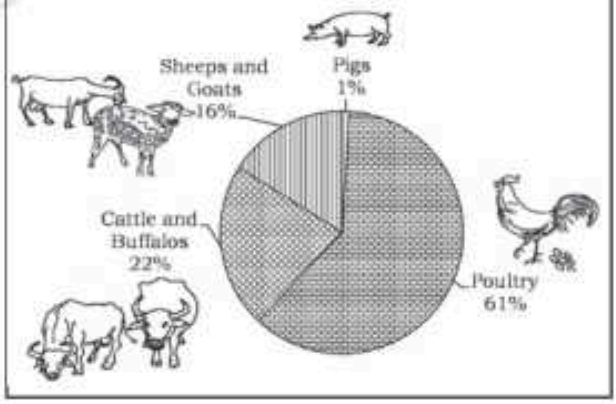
Tamil Nadu Women in Agriculture (TANWA) was a project initiated in the late 1980s in Tamil Nadu to train women in latest agricultural techniques and in organic farming. It encouraged women to actively participate in raising agricultural productivity and family income. At a Farm Women's Group in Thiruchirapalli, run by Anthoniammal, trained women are successfully making and selling vermicompost and earning money from this venture. Many other Farm Women's Groups are creating savings in their group by functioning like mini banks through a micro-credit system. With the accumulated savings, they promote small-scale household activities like mushroom cultivation, soap manufacture, doll making or other income-generating activities. There could be similar initiatives in your area or your neighborhood rural areas. Talk to organisation or women entrepreneurs and discuss the details in the class.

As agriculture is already overcrowded, a major proportion of the increasing labour force needs to find alternate employment opportunities in other non-farm sectors. Non-farm economy has several segments in it; some possess dynamic linkages that permit healthy growth while others are in subsistence, low productivity propositions. The dynamic sub-sectors include **agro-processing industries,** food processing industries, leather industry, tourism, etc. Those sectors which have the potential but seriously lack infrastructure and other support include traditional home-based industries like pottery, crafts, handlooms etc. Majority of rural women find employment in agriculture while men generally look for non-farm employment. In recent times, women have also begun looking for non-farm jobs (see Box 5.2).

Animal Husbandry: In India, the farming community uses the mixed crop-livestock farming system —cattle, goats, fowl are the widely held species. Livestock production provides increased stability in income, food security, transport, fuel and nutrition for the family without disrupting other food-producing activities. Today, livestock sector alone provides alternate livelihood options to over 70 million small and marginal farmers including landless labourers. A significant number of women also find employment in the livestock sector.

Chart 5.1 shows the distribution of livestock in India. Poultry accounts for the largest share with 61 per cent

Chart 5.1: Distribution of Poultry and Livestock in India, 2019



Reprint 2026-27



Fig. 5.3 *Sheep rearing — an important income augmenting activity in rural areas*

followed by others. Other animals which include camels, asses, horses, ponies and mules are in the lowest rung. India had about 303 million cattle, including 110 million buffaloes in 2019. Performance of the Indian dairy sector over the last three decades has been quite impressive. Milk production in the country has increased by about twelve times between 1951-2021. This can be attributed mainly to the successful implementation of 'Operation Flood'. It is a system whereby all the farmers can pool their milk produced according to different grading (based on quality), processed and marketed to urban centres through cooperatives. In this system the farmers are assured of a fair price and income from the supply of milk to urban markets. As pointed out earlier Gujarat state is held as a success story in the efficient implementation of milk cooperatives which has been emulated by many states. Gujarat, Madhya Pradesh, Uttar Pradesh, Andhra Pradesh, Maharashtra, Punjab and Rajasthan, are major milk producing states. Meat, eggs, wool and other by-products are also emerging as important productive sectors for diversification.

Fisheries: The fishing community regards the water body as 'mother' or 'provider'. The water bodies consisting of sea, oceans, rivers, lakes, natural aquatic ponds, streams etc. are, therefore, an integral and life-giving source for the fishing community. In India, after progressive increase in budgetary allocations and introduction of new technologies in fisheries and aquaculture, the development of fisheries has come a long way. Presently, fish production from inland sources contributes about 75 per cent to the total value of fish production and the balance 25 per cent comes from the marine sector (sea and oceans). Today total fish production accounts for about 1.5 per cent of the total GDP. In India, West Bengal, Andhra Pradesh, Kerala, Gujarat, Maharashtra and Tamil Nadu are major fish producing states. A large share of fishworker families are poor. Rampant underemployment, low per capita earnings, absence of mobility of labour to other sectors and a high rate of illiteracy and indebtedness are some of the major problems fishing community face today. Even though women are not involved in active fishing, about 60 per cent of the workforce in export marketing and 40 per cent in internal marketing are women. There is a need to increase credit facilities through cooperatives and SHGs for

Reprint 2026-27

fisherwomen to meet the working capital requirements for marketing.

Horticulture: Blessed with a varying climate and soil conditions, India has adopted growing of diverse horticultural crops such as fruits, vegetables, tuber crops, flowers, medicinal and aromatic plants, spices and plantation crops. These crops play a vital role in providing food and nutrition, besides addressing employment concerns. Horticulture sector contributes nearly one-third of the value of agriculture output and six per cent of Gross Domestic Product of India. India has emerged as a world leader in producing a variety of fruits like mangoes, bananas, coconuts, cashew nuts and a number of spices and is the second largest producer of fruits and vegetables. Economic condition of many farmers engaged in horticulture has improved and it has become a means of improving livelihood for many unprivileged classes. Flower harvesting, nursery maintenance,



Fig. 5.5 *Women in rural households take up bee-keeping as an entrepreneurial activity*

hybrid seed production and tissue culture, propagation of fruits and flowers and food processing are highly remunerative employment options for women in rural areas.

Though, in terms of numbers, our livestock population is quite impressive but its productivity is quite low as compared to other countries. It requires improved technology and promotion of good breeds of animals to enhance productivity. Improved veterinary care and credit facilities to small and marginal farmers and landless labourers would enhance sustainable livelihood options through livestock production. Production of fisheries has already increased substantially.

However problems related to over- fishing and pollution need to be regulated



Fig. 5.4 *Poultry has the largest share of total livestock in India*

Reprint 2026-27

and controlled. Welfare programmes for the fishing community have to be reoriented in a manner which can provide long-term gains and sustenance of livelihoods. Horticulture has emerged as a successful sustainable livelihood option and needs to be encouraged significantly. Enhancing its role requires investment in infrastructure like electricity, cold storage systems, marketing linkages, small-scale processing units and technology improvement and dissemination.

Other Alternate Livelihood Options: The IT has revolutionised many sectors in the Indian economy. There is broad consensus that IT can play a critical role in achieving sustainable development and food security in the twenty-first century. Governments can predict areas of food insecurity and vulnerability using appropriate information and software tools so that action can be taken to prevent or reduce the likelihood of an emergency. It also has a positive impact on the agriculture sector as it can disseminate information regarding emerging technologies and its applications, prices, weather and soil conditions for growing different crops etc. Though IT is, by itself, no catalyst of change but it can act as a tool for releasing the creative potential and knowledge embedded in the society. It also has potential of employment generation in rural areas. Experiments with IT and its application to rural development are carried out in different parts of India.

5.6 Sustainable Development and Organic Farming

In recent years, awareness of the harmful effect of chemical-based fertilisers and pesticides on our health is on a rise. Conventional agriculture relies heavily on chemical fertilisers and toxic pesticides etc., which enter the food supply, penetrate the water sources, harm the livestock, deplete the soil and devastate natural eco-systems. Efforts in evolving technologies which are eco-friendly are essential for sustainable development and one such technology which is eco-friendly is organic farming. In short, organic agriculture is a whole system of

Box 5.3: Adoption of Village by Parliamentarians

In October, 2014, The Government of India introduced a new scheme called Saansad Adarsh Gram Yojana (SAGY). Under this scheme, Members of India's Parliament need to identify and develop one village from their constituencies. To begin with, MPs can develop one village as a model village by 2016, and two more by 2019, covering over 2,500 villages in India. According to the scheme, the village can have a population of 3,000-5,000 in the plains and 1,000-3,000 in the hills and should not be MPs' own or their spouse's village. MPs are expected to facilitate a village development plan, motivate villagers to take up activities and built infrastructure in the areas of health, nutrition and education.

Source: *www.pib.nic.in accessed on 24 October 2014.*

Reprint 2026-27

Box 5.4: Organic Food

Organic food is growing in popularity across the world. Many countries have around 10 per cent of their food system under organic farming. There are many retail chains and supermarkets which are accorded with **green status** to sell organic food. Moreover, organic foods command higher price of around 10-100 per cent than conventional ones.

farming that restores, maintains and enhances the ecological balance. There is an increasing demand for organically grown food to enhance food safety throughout the world (see Box 5.4).

Benefits of Organic Farming: Organic agriculture offers a means to substitute costlier agricultural inputs (such as HYV seeds, chemical fertilisers, pesticides etc.) with locally produced organic inputs that are cheaper and thereby generate good returns on investment. Organic agriculture also generates income through exports as the demand for organically grown crops is on a rise. Studies across countries have shown that organically grown food has more nutritional value than chemical farming thus providing us with healthy foods. Since organic farming requires more labour input than conventional farming, India will find organic farming an attractive proposition. Finally, the produce is pesticide-free and produced in an environmentally sustainable way (see Box 5.5).

Popularising organic farming requires awareness and willingness on the part of farmers to adapt to new technology. Inadequate infrastructure and the problem of marketing the products are major concerns which need to be addressed apart from an

Box 5.5: Organically Produced Cotton in Maharashtra

In 1995, when Kisan Mehta of *Prakruti* (an NGO) first suggested that cotton, the biggest user of chemical pesticides, could be grown organically, the then Director of the Central Institute for Cotton Research, Nagpur, famously remarked, "Do you want India to go naked?" At present, as many as 130 farmers have committed 1,200 hectares of land to grow cotton organically on the International Federation of Organic Agriculture Movement's standards. The produce was later tested by the German Accredited Agency, AGRECO, and found to be of high quality. Kisan Mehta feels that about 78 per cent of Indian farmers are marginal farmers owning about less than 0.8 hectare but accounting for 20 per cent of India's cultivable land. For such farmers, organic agriculture is more profitable in terms of money and soil conservation in the long run. Visit a farm in your locality which uses organic manure and discuss the uses associated with the farming practices. Present the report in your class.

Source: *Lyla Bavadam, A Green Alternative, Frontline, 29 July 2005.*

Reprint 2026-27

appropriate agriculture policy to promote organic farming. It has been observed that the yields from organic farming are less than modern agricultural farming in the initial years. Therefore, small and marginal farmers may find it difficult to adapt to large-scale production. Organic produce may also have more blemishes and a shorter shelf life than sprayed produce. Moreover choice in production of off-season crops is quite limited in organic farming. Nevertheless, organic farming helps in sustainable development of agriculture and India has a clear advantage in producing organic products for both domestic and international markets. Do you think food and non-food items cultivated using organic farming methods will be cheaper?

5.7 Conclusion

It is clear that until and unless some spectacular changes occur, the rural sector might continue to remain backward. There is a greater need today to make rural areas more vibrant through diversification into dairying, poultry, fisheries, vegetables and fruits and linking up the rural production centres with the urban and foreign (export) markets to realise higher returns on the investments for the products. Moreover, infrastructure elements like credit and marketing, farmer-friendly agricultural policies and a constant appraisal and dialogue between farmers' groups and state agricultural departments are essential to realise the full potential of the sector.

Today we cannot look at the environment and rural development as two distinct subjects. There is need to invent or procure alternate sets of eco-friendly technologies that lead to sustainable development in different circumstances. From these, each rural



Work These Out

- Make a list of five popular items that are organically produced in India.
- Visit a nearby super market, vegetable shop and/or a departmental shop. Identify a few products. Prepare a chart comparing a few goods that are produced organically and in the normal way on the basis of their prices, shelf life, quality and the kind of advertisement through which they are popularised.
- Visit a horticultural farm in the nearby locality. Collect the details of goods that they cultivate on the farm. They could have diversified their cropping patterns. Discuss with them the merits and demerits of the diversification.

© NCERT not to be republished

Reprint 2026-27

6.2 Workers and Employment

What is employment? Who is a worker? When a farmer works on fields, he or she produces food grains and raw materials for industries. Cotton becomes cloth in textile mills and in powerlooms. Lorries transport goods from one place to another. We know that the total money value of all such final goods and services produced in a country in a year is called its gross domestic product for that year. When we also consider what we pay for our imports and get from our exports we find that there is a net earning for the country which may be positive (if we have exported more in value terms than imported) or negative (if imports exceeded exports in value terms) or zero (if exports and imports were of the same value). When we add this earning (plus or minus) from foreign transactions, what we get is called the country's gross national product for that year.

Those activities which contribute to the gross national product are called economic activities. All those who are engaged in economic activities, in whatever capacity — high or low, are workers. Even if some of them temporarily abstain from work due to illness, injury or other physical disability, bad weather, festivals, social or religious functions, they are also workers. Workers also include all those who help the main workers in these activities. We generally think of only those who are paid by an employer for their work as workers. This is not so. Those who are self-employed are also workers.

The nature of employment in India is multifaceted. Some get employment throughout the year; some others get employed for only a few months in a year. Many workers do not get fair wages for their work. While estimating the number of workers, all those who are engaged in economic activities are included as employed. You might be interested in knowing the number of people actively engaged in various economic activities. During 2022-23, India had a workforce of about 545 million. Since majority of our people reside in rural areas, the proportion of workforce residing there is higher. The



Work This Out

In your house or neighbourhood, you might come across many women who, even though they have technical degrees and diplomas and also free time to go to work, do not go to work. Ask them the reasons for not going to work. List all of them and discuss in the classroom whether they should go for work and why, and also ways by which they could be sent for work. Some social scientists argue that housewives working at home without getting paid for that work must also be regarded as contributing to the gross national product and therefore, as engaged in an economic activity. Would you agree?

Reprint 2026-27

rural workers constitute about two-thirds of this 545 million. Men form the majority of workforce in India. About 77 per cent of the workers are men and the rest are women (men and women include child labourers in respective sexes). Women workers account for one-fourth of the rural workforce whereas in urban areas, they are just one-fifth of the workforce. Women carry out works like cooking, fetching water and fuelwood and participate in farm labour. They are not paid wages in cash or in the form of grains; at times they are not paid at all. For this reason, these women are not categorised as workers. Economists argue that these women should also be called workers. What do you think?

6.3 Participation of People in Employment

Worker-population ratio is an indicator which is used for analysing the employment situation in the country. This ratio is useful in knowing the proportion of population that is actively contributing to the production of goods and services of a country. If the ratio is higher, it means that the engagement of people is greater; if the ratio for a country is medium, or low, it means that a very high proportion of its population is not involved directly in economic activities.

You might have already studied, in lower classes, the meaning of the term 'population'. Population is defined as the total number of people who reside in a particular locality at a particular point of time. If you want to know the worker-population ratio for India, divide the total number of workers in India by the population in India and multiply it by 100, you will get the worker-population ratio for India.

If you look at Table 6.1, it shows the different levels of participation of people in economic activities. For every 100 persons, about 41 (by rounding off 41.1) are workers in India. In urban areas, the proportion is about 38, whereas in rural India, the ratio is about 42. Why is there such a difference? People in rural areas have limited resources to earn a higher income and participate more in the employment market. Many do not go to schools, colleges and other training institutions. Even if some go, they discontinue in the middle to join the workforce; whereas, in urban areas, a considerable section is able to study in various educational institutions. Urban people have a variety of employment opportunities. They look for the appropriate job to suit their qualifications and skills. In rural areas, people cannot stay at home as their economic condition may not allow them to do so.

TABLE 6.1

Worker-Population Ratio in India, 2023-24

Sex	*Worker-Population Ratio*		
	Total	*Rural*	*Urban*
Men	56.4	56.3	56.4
Women	30.7	34.8	20.7
Total	**43.7**	**45.6**	**38.9**

Reprint 2026-27

Work These Out

- Any study of employment must start with a review of the worker-population ratios — why?
- In some communities, you might have noticed that even if the males do not earn a high income, they do not send women to work. Why?

Compared to females, more males are found to be working. The difference in participation rates is very large in urban areas: for every 100 urban females, only about 21 are engaged in some economic activities. In rural areas, for every 100 rural women about 35 participate in the employment market. Why are women, in general, and urban women, in particular, not working? It is common to find that where men are able to earn high incomes, families discourage female members from taking up jobs.

Going back to what has already been mentioned above, many household activities done by women are not recognised as productive work. This narrow definition of work leads to non-recognition of women's work and, therefore, to the underestimation of the number of women workers in the country. Think of the women actively engaged in many activities within the house and at family farms who are not paid for such work. As they certainly contribute to the maintenance of the household and farms, do you think that their number should be added to the number of women workers?

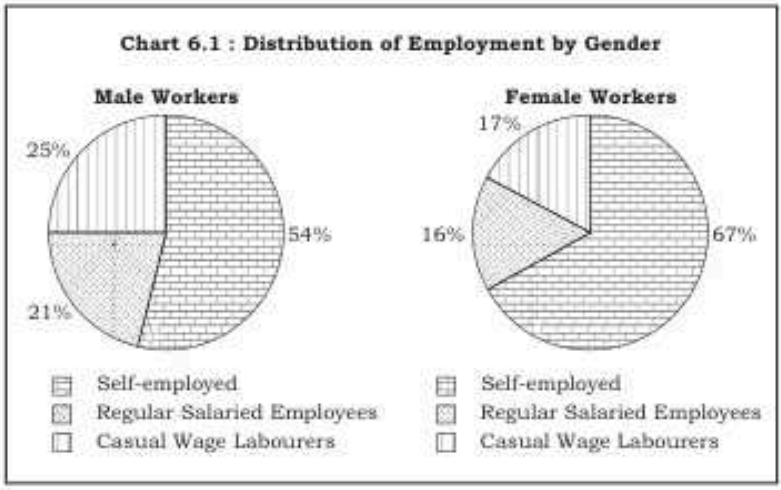
6.4 Self-employed and Hired Workers

Does the worker-population ratio say anything about workers' status in society or about the working conditions? By knowing the status with which a worker is placed in an enterprise, it may be possible to know one dimension — quality of employment in a country. It also



Fig. 6.2 *Brick-making: a form of casual work*

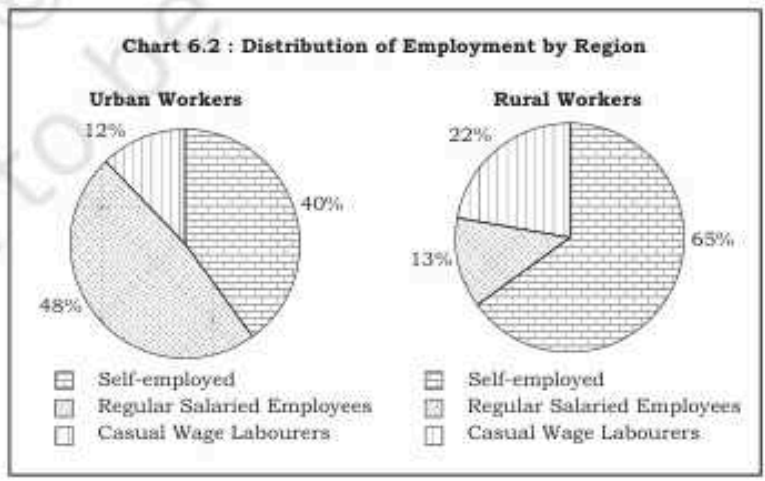
Reprint 2026-27



enables us to know the attachment a worker has with his or her job and the authority she or he has over the enterprise and over other co-workers.

Let us take three workers from the construction industry — a cement shop owner, a construction worker and a civil engineer of a construction company. Since the status of each one of them is different from another, they are also called differently. Workers who own and operate an enterprise to earn their livelihood are known as **self-employed**. Thus the cement shop owner is self-employed. About 58 per cent workforce in India belongs to this category. The construction workers are known as **casual wage labourers**; they account for about 20 per cent of India's workforce. Such labourers are casually engaged in others' farms and, in return, get a remuneration for the work done. Workers like the civil engineer working in the construction company account for 22 per cent of India's workforce. When a worker is engaged by someone or an enterprise and paid his or her wages on a regular basis, they are known as **regular salaried employees** (see table 6.3).

Look at Chart 6.1 you will notice that self-employment is a major source of livelihood for both men and women as this category accounts for more than 50 per cent of the workforce. Casual wage work is the second major source for women. When it comes to regular



Reprint 2026-27

salaried employment, men are found to be so engaged in greater proportion. Men form 21 per cent whereas women form 16 per cent. The gap between men and women is very less.

When we compare the distribution of workforce in rural and urban areas in Chart 6.2 you will notice that the self-employed are found more in rural areas than in urban areas. In the latter, both self-employment and regular wage salaried jobs are greater. In rural areas, since majority of those depending on farming own plots of land and cultivate independently, the share of self-employed is greater.

The nature of work in urban areas is different. Obviously everyone cannot run factories, shops and offices of various types. Moreover enterprises in urban areas require workers on a regular basis.



Work These Out

- We generally think that only those who are doing paid work regularly or casually such as agricultural labourers, factory workers, those who work in banks and other offices as assistants and clerks are workers. From the above discussion, you must have understood that those who are self-employed such as pavement vegetable vendors, professionals such as lawyers, doctors and engineers are also workers. Mark (a), (b) and (c) against self-employed, regular salaried employees and casual wage labourers respectively:
  1. Owner of a saloon
  2. Worker in a rice mill who is paid on daily basis but employed regularly
  3. Cashier in State Bank of India
  4. Typist working in a state government office on a daily wage basis but paid monthly
  5. A handloom weaver
  6. Loading worker in wholesale vegetable shop
  7. Owner of a cold drinks shop which sells Pepsi, Coca Cola and Mirinda
  8. Nurse in a private hospital who gets monthly salary and has been working regularly for the past 5 years.
- Economists point out that casual wage labourers are the most vulnerable among the three categories. Could you locate who these workers are and where they are found and why?
- Can we say that the self-employed earn more than the casual wage labourers or regular salaried employees? Identify a few other indicators of quality of employment.

© NCERT not to be republished

6.5 Employment in Firms, Factories and Offices



Fig. 6.3 *Garment workers: upcoming factory employment for women*

In the course of economic development of a country, labour flows from agriculture and other related activities to industry and services. In this process, workers migrate from rural to urban areas. Eventually, at a much later stage, the industrial sector begins to lose its share of total employment as the service sector enters a period of rapid expansion. This shift can be understood by looking at the distribution of workers by industry. Generally, we divide all economic activities into eight different industrial divisions. They are (i) Agriculture (ii) Mining and Quarrying (iii) Manufacturing (iv) Electricity, Gas and Water Supply (v) Construction (vi) Trade (vii) Transport and Storage and (viii) Services. For simplicity, all the working persons engaged in these divisions can be clubbed into three major sectors viz., (a) primary sector which includes (i) and (ii), (b) secondary sector which includes (iii), (iv) and (v) and (c) service sector which includes divisions (vi), (vii) and (viii). Table 6.2 shows the distribution of working persons in different industries during the year 2023-24. Secondary and Tertiary/Service sectors could also be called as non-farm sectors.

Primary sector is the main source of employment for majority of workers

TABLE 6.2

Distribution of Workforce by Industry, 2023-24

Industrial Category	*Place of Residence*		*Sex*		*Total*
	Rural	*Urban*	*Male*	*Female*	
Primary Sector	59.8	6.7	36.3	64.4	46.1
Secondary Sector	21.4	32.4	28.8	15.6	24.1
Tertiary / Service Sector	18.8*	60.9*	34.9*	20.0*	29.8*
Total	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**

* Rounded off data

Reprint 2026-27

Work This Out

- All newspapers have one section meant for job opportunities. Some also devote an entire supplement in a day or every week like **Opportunities** in *The Hindu* or **Ascent** in The *Times of India*. Many companies advertise vacancies for various positions. Cut those sections. Develop a table which contains four columns: whether the company is private or public, name of the post, number of posts, sector — primary, secondary or tertiary — and qualification required. Analyse the table in the classroom about jobs advertised in the newspapers.

in India. Secondary sector provides employment to only about 24 per cent of workforce. About 30 per cent of workers are in the service sector. Table 6.2 also shows that about 60 per cent of the workforce in rural India depends on agriculture, forestry and fishing. About 21 per cent of rural workers are working in manufacturing industries, construction and other industrial activities. Service sector provides employment to about 19 per cent of rural workers. Agriculture is not a major source of employment in urban areas where people are mainly engaged in the service sector. About 61 per cent of urban workers are in the service sector. The secondary sector gives employment to about one-third of urban workforce.

Though both men and women workers are concentrated in the primary sector, women workers' concentration is very high there. About 64 per cent of the female workforce is employed in the primary sector whereas 36 per cent of males work in that sector. Men get opportunities in both secondary and service sectors.

6.6 Growth and Changing Structure of Employment

In Chapters 2 and 3, you might have studied about the planning strategies in detail. Here we will look at two developmental indicators — growth of employment and GDP. Nearly seventy years of planned development have been aimed at expansion of the economy through increase in national output and employment.

During the period 1951-2023, Gross Domestic Product (GDP) of India grew positively and was higher than the employment growth. However, there was always fluctuation in the growth of GDP. During this period, employment grew at the rate of not more than 2 per cent most of the times. Only during two points of time (2022-2023 and 1999-2005), over the period of seven decades employment growth went beyond 2 per cent.

Chart 6.3 also points at another disheartening development in the late 1990s: employment growth started declining and reached the level of growth that India had in the early stages of planning. During these years, we also find a widening gap

Reprint 2026-27

between the growth of GDP and employment. This means that in the Indian economy, without generating employment, we have been able to produce more goods and services. Scholars refer to this phenomenon as jobless growth.

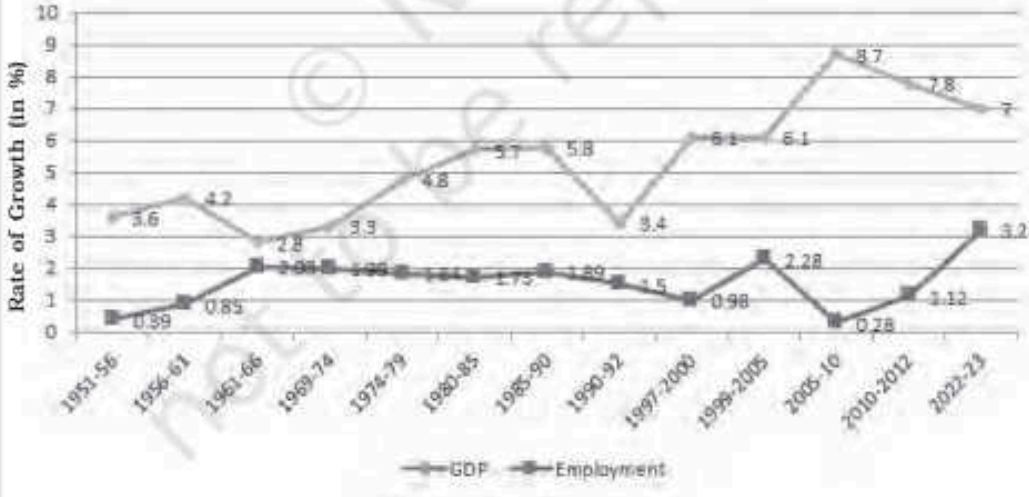
So far we have seen how employment has grown in comparison to GDP. Now it is necessary to know how the growth pattern of employment and GDP affected different sections of workforce. From this we will also be able to understand what types of employment are generated in our country.

Let us look at two indicators that we have seen in the preceding sections — employment of people in various industries and their status. We know that India is an agrarian nation; a major section of population lives in rural areas and is dependent on agriculture as their main livelihood. Developmental strategies in many countries, including India, have aimed at reducing the proportion of people depending on agriculture.

Distribution of workforce by industrial sectors shows substantial shift from farm work to non-farm work (see Table 6.3). In 1972-73, about 74 per cent of workforce was engaged in primary sector and in 2023-24, this proportion has declined to about 46 per cent. Secondary and service sectors are showing promising future for the Indian workforce. You may notice that the shares of these sectors have increased from 11 to 24 per cent and 15 to 30 per cent, respectively.

The distribution of workforce in different status indicates that over the

Chart 6.3: Growth of Employment and Gross Domestic Product, 1951–2023 (%)



Note: **This is the period for which comparable and authentic data are available.*

Reprint 2026-27

TABLE 6.3

Trends in Employment Pattern (Sector-wise and Status-wise), 1972-2024 (in %)

Item	*1972-73*	*1983*	*1993-94*	*2011-2012*	*2017-2018*	*2023–24*
Sector						
Primary	74.3	68.6	64	48.9	46.1	46.1
Secondary	10.9	11.5	16	24.3	24.9	24.1
Services	14.8	16.9	20	26.8	29.0	29.8*
Total	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**
Status						
Self-employed	61.4	57.3	54.6	52.0	55.6	58.4
Regular Salaried Employees	15.4	13.8	13.6	18.0	21.1	21.7
Casual Wage Labourers	23.2	28.9	31.8	30.0	23.3	19.9*
Total	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**

* Rounded off data

last five decades (1972-2024), people have moved from self-employment and regular salaried employment to casual wage work. Yet self-employment continues to be the major employment provider. Look at the last column of table 6.3. How do you understand the stagnation of secondary sector and moderate rise in self-employement during 2011-24? Discuss in the class. Scholars call the process of moving from self-employment and regular salaried employment to casual wage work as **casualisation of workforce**. This makes the workers highly vulnerable. How? Look at the case study of Ahmedabad in the preceding section.

In the mean time you also notice in 2017-24 a moderate rise in the share of regular salaried employees. How do you explain this phenomenon?



Work These Out

- Do you know that maintaining employment growth at 2 per cent for a country like India is not an easy thing? Why?
- What will happen if there is no additional employment generated in the economy even though we are able to produce goods and services in the economy? How could jobless growth happen?
- Economists say that if casualisation increases the earning of the people, such phenomenon should be welcomed. Suppose a marginal farmer becomes a full-time agricultural labourer, do you think he will be happy even if he earns more in his daily wage work? Or will a permanent and regular worker of the pharmaceutical industry be happy if he becomes a daily wage labourer, even if his or her overall earnings increase? Discuss in the classroom.

Reprint 2026-27

6.7 Informalisation of Indian Workforce

In the previous section we have found that the proportion of hired work jobs - working for others has been increasing. One of the objectives of development planning in India, since India's independence, has been to provide decent livelihood to its people. It has been envisaged that the industrialisation strategy would bring surplus workers from agriculture to industry with better standard of living as in developed countries. We have seen in the preceding section, that even after 70 years of planned development, more than half of the Indian workforce depends on farming as the major source of livelihood.

Economists argue that, over the years, the quality of employment has been deteriorating. Even after working for more than 10-20 years, why do some workers not get maternity benefit, provident fund, gratuity and pension? Why does a person working in the private sector get a lower salary as compared to another person doing the same work but in the public sector?

You may find that a small section of Indian workforce is getting regular income. The government, through its **labour laws**, enable them to protect their rights in various ways. This section of the workforce forms **trade unions**, bargains with employers for better wages and other **social security** measures. Who are they? To know this we classify workforce into two categories: workers in formal and informal sectors, which are also referred to as organised and unorganised sectors. All the **public sector establishments** and those **private sector establishments** which employ 10 hired workers or more are called formal sector establishments and those who work in such establishments are formal sector workers. All other enterprises and workers working in those enterprises form the informal sector. Thus, informal sector includes millions of farmers, agricultural labourers, owners of small enterprises and people working in those enterprises as also the self-employed who do not have any hired workers. It also includes all non-farm casual wage labourers who work for more than one employer such as construction workers and headload workers. You may note that this is one of the ways of classifying workers. There could be other ways of classification as well. Discuss the possible ways in the class.

Box 6.1: Formal Sector Employment

The information relating to employment in the formal sector is collected by the Union Ministry of Labour through employment exchanges located in different parts of the country. Do you know who is the major employer in the formal sector in India? In 2012, out of about 30 million formal sector workers, about 18 million workers were employed by the public sector. Here also men form the majority, as women constitute only about one-sixth of the formal sector workforce. Economists point out that the reform process initiated in the early 1990s resulted in a decline in the number of workers employed in the formal sector. What do you think?

Reprint 2026-27

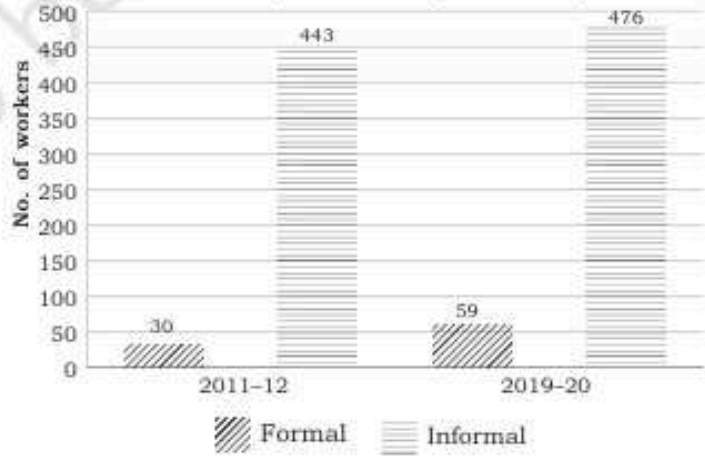


Fig. 6.4 *Road side vending: an increasing variety of informal sector employment*

Those who are working in the formal sector enjoy social security benefits. They earn more than those in the informal sector. Developmental planning envisaged that as the economy grows, more and more workers would become formal sector workers and the proportion of workers engaged in the informal sector would dwindle. But what has happened in India? Look at the following chart which gives the distribution of workforce in formal and informal sectors.

In 2019–2020 there were about 535 million workers in India. There were about 59 million workers in the formal sector (Chart 6.4). Can you estimate the percentage of people employed in the formal sectors in the country? About only 11 per cent (59/535 100)! Thus, the rest 89 per cent are in the informal sector. In 2011–2012, the year for which gender wise data on formal-informal sector employment is available about 20 per cent of formal sector and 30 per cent of informal sector workers were women.

Chart 6.4: Workers in Formal and Informal Sector, 2011–2020 (in millions)



Reprint 2026-27

Box 6.2: Informalisation in Ahmedabad

Ahmedabad is a prosperous city with its wealth based on the produce of more than 60 textile mills with a labour force of 1,50,000 workers employed in them. These workers had, over the course of the century, acquired a certain degree of income security. They had secure jobs with a living wage; they were covered by social security schemes protecting their health and old age. They had a strong trade union which not only represented them in disputes but also ran activities for the welfare of workers and their families. In the early 1980s, textile mills all over the country began to close down. In some places, such as Mumbai, the mills closed rapidly. In Ahmedabad, the process of closure was long drawn out and spread over 10 years. Over this period, approximately over 80,000 permanent workers and over 50,000 non-permanent workers lost their jobs and were driven to the informal sector. The city experienced an economic recession and public disturbances, especially communal riots. A whole class of workers was thrown back from the middle class into the informal sector, into poverty. There was widespread alcoholism and suicides, children were withdrawn from school and sent to work.



Change in the balance of power in a house: an unemployed mill worker peeling garlic whereas his wife has a new job of beedi rolling.

Source: *Renana Jhabvala, Ratna M. Sudarshan and Jeemol Unni (Ed.) Informal Economy at Centre Stage: New Structures of Employment, Sage Publications, New Delhi, 2003, pp.265.*

Since the late 1970s, many developing countries, including India, started paying attention to enterprises and workers in the informal sector as employment in the formal sector is not growing. Workers and enterprises in the informal sector do not get regular income; they do not have any protection or regulation from the government. Workers are dismissed without any compensation. Technology used in the informal sector enterprises is outdated; they also do not maintain any accounts. Workers of this sector live in slums and are squatters. Of late, owing to the efforts of the International Labour Organisation (ILO), the Indian government has

Reprint 2026-27

Work These Out

Tick (✓) mark against those which are in the informal sector

- Worker in a hotel which has seven hired workers and three family workers
- A private school teacher in a school which has 25 teachers
- A police constable
- Nurse in a government hospital
- Cycle-rickshaw puller
- The owner of a textile shop employing nine workers
- Driver of a bus company which has more than 10 buses with 20 drivers, conductors and other workers
- Civil engineer working in a construction company which has 10 workers
- Computer operator in the state government office working on a temporary basis
- A clerk in the electricity office.

initiated the modernisation of informal sector enterprises and provision of social security measures to informal sector workers.

6.8 Unemployment

You might have seen people looking for jobs in newspapers. Some look for a job through friends and relatives. In many cities, you might find people standing in some select areas looking for people to employ them for that day's work. Some go to factories and offices and give their bio-data and ask whether there is any vacancy in their factory or office. Many in the rural areas do not go out and ask for a job but stay home when there is no work. Some go to



Fig. 6.5 *Unemployed mill workers waiting for casual jobs*

Reprint 2026-27



Fig. 6.6 *Sugar cane cutters: disguised unemployment is common in farm works*

employment exchanges and register themselves for vacancies notified through employment exchanges. The National Statistical Office (Previously it was known as National Sample Survey Organisation) defines unemployment as a situation in which all those who, owing to lack of work, are not working but either seek work through employment exchanges, intermediaries, friends or relatives or by making applications to prospective employers or express their willingness or availability for work under the prevailing condition of work and remunerations. There are a variety of ways by which an unemployed person is identified. Economists define unemployed person as one who is not able to get employment of even one hour in half a day.

There are three sources of data on unemployment : Reports of Census of India, National Statistical Office's Reports of Employment and Unemployment Situation, Annual Reports of Periodic Labour Force Survey, and Directorate General of Employment and Training data of Registration with Employment Exchanges. Though they provide different estimates of unemployment, they do provide us with the attributes of the unemployed and the variety of unemployment prevailing in our country.

Do we have different types of unemployment in our economy? The situation described in the first paragraph of this section is called open unemployment. Economists call unemployment prevailing in Indian farms as disguised unemployment. What is disguised unemployment? Suppose a farmer has four acres of land and he actually needs only two workers and himself to carry out various operations on his farm in a year, but if he employs five workers and his family members such as his wife and children, this situation is known as disguised unemployment. One study conducted in the late 1950s showed about one-third of agriculture workers in India as disguisedly unemployed.

You may have noticed that many people migrate to an urban area, pick up a job and stay there for some time, but come back to their home villages as soon as the rainy season begins. Why do they do so? This is because work in agriculture is seasonal; there are no employment opportunities in the village for all months in the year. When there is no work to do on farms, people go to urban areas and look for jobs.

Reprint 2026-27

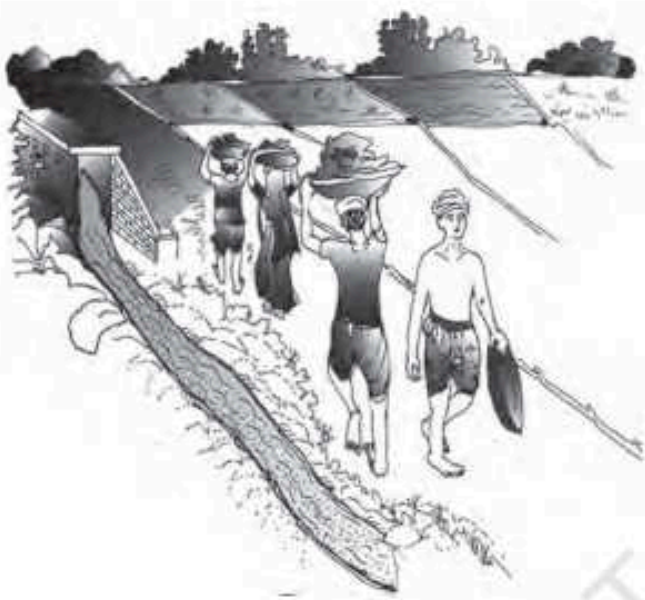


Fig. 6.7 *Dam construction work is a direct way of employment generation by the government*

This kind of unemployment is known as seasonal unemployment. This is also a common form of unemployment prevailing in India.

Though we have witnessed slow growth of employment, have you seen people being unemployed over a very long time? Scholars say that in India, people cannot remain completely unemployed for very long because their desperate economic condition would not allow them to be so. You will rather find them being forced to accept jobs that nobody else would do, unpleasant or even dangerous jobs in unclean, or unhealthy surroundings. The Central and State governments take initiatives and generate employment to facilitate a decent living for low income families through various measures. These will be discussed in the following section.

6.9 Government and Employment Generation

You may recall about the Mahatma Gandhi National Rural Employment Guarantee Act 2005. It promises 100 days of guaranteed wage employment to all rural households who volunteer to do unskilled manual work. The scheme based on this Act is one of the many measures the government has implemented to generate employment for those who are in need of jobs in rural areas.

Since Independence, the Union and State governments have played an important role in generating employment or creating opportunities for employment generation. Their efforts can be broadly categorised into two — direct and indirect. In the first category, as you have seen in the preceding section, the government employs people in various departments for administrative purposes. It also runs industries, hotels and transport companies, and hence, provides employment directly to workers. When the output of goods and services from government enterprises increases, then private enterprises which receive raw materials from government enterprises will also raise their output and hence increase the number of employment opportunities in the economy. For example, when a government owned

Reprint 2026-27